



ENHANCING DOCUMENT SECURITY: A DUAL APPROACH WITH NLP AND CRYPTOGRAPHIC TECHNIQUES



A PROJECT REPORT

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*In partial fulfilment for the award of the degree
of*

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING

DHIRAJLAL GANDHI COLLEGE OF TECHNOLOGY

(An Autonomous Institution)

SALEM – 636 309

ANNA UNIVERSITY: CHENNAI – 600 025

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BONAFIDE CERTIFICATE

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INTERNAL EXAMINER

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ABSTRACT

In today's digital era, the risk of document theft and unauthorized access to sensitive information has become increasingly prevalent, posing serious threats to individuals, organizations, and governments. Traditional security measures such as access control and basic encryption often fall short in addressing these challenges, especially with the growing complexity of cyber threats. This project proposes a dual-layered defense system that integrates Natural Language Processing (NLP) and Cryptography to enhance document security. NLP techniques are utilized to analyze, classify, and monitor document content in real-time, enabling the system to identify sensitive data, generate unique document fingerprints, and detect unauthorized changes or content extraction that traditional systems might miss. To strengthen protection further, the system incorporates Elliptical curve cryptography encryption to ensure that documents remain secure during storage and transmission, making them unreadable to unauthorized users. Additionally, the project includes a deception strategy by generating realistic fake documents and storing them in alternative repositories to mislead potential attackers. This hybrid approach of combining content-based monitoring, strong encryption, and decoy deployment creates a comprehensive and proactive solution for securing sensitive information. It offers a significant advancement in combating document theft and unauthorized modifications, making it highly applicable across industries that prioritize data confidentiality and integrity. Future enhancements include AI-based threat detection, blockchain-based document logging, secure cloud storage, support for multiple file formats, and a mobile app with biometric authentication for real-time access and alerts. This hybrid approach provides a comprehensive and proactive solution for safeguarding sensitive information across industries.

திட்டப்பணிசுருக்கம்

இன்றைய டிஜிட்டல் சகாப்தத்தில், ஆவணத் திருட்டு மற்றும் முக்கியமான தகவல்களை அங்கீகரிக்கப்படாத முறையில் அணுகுவதற்கான ஆபத்து அதிகரித்து வருகிறது, இது தனிநபர்கள், நிறுவனங்கள் மற்றும் அரசாங்கங்களுக்கு கடுமையான அச்சுறுத்தல்களை ஏற்படுத்துகிறது. அணுகல் கட்டுப்பாடு மற்றும் அடிப்படை குறியாக்கம் போன்ற பாரம்பரிய பாதுகாப்பு நடவடிக்கைகள் பெரும்பாலும் இந்த சவால்களை எதிர்கொள்வதில் தோல்வியடைகின்றன, குறிப்பாக சைபர் அச்சுறுத்தல்களின் வளர்ந்து வரும் சிக்கலான தன்மையுடன். ஆவணப் பாதுகாப்பை மேம்படுத்த இயற்கை மொழி செயலாக்கம் (NLP) மற்றும் கிரிப்டோகிராஃபி ஆகியவற்றை ஒருங்கிணைக்கும் இரட்டை அடுக்கு பாதுகாப்பு அமைப்பை இந்தத் திட்டம் முன்மொழிகிறது. ஆவண உள்ளடக்கத்தை நிகழ்நேரத்தில் பகுப்பாய்வு செய்ய, வகைப்படுத்த மற்றும் கண்காணிக்க NLP நுட்பங்கள் பயன்படுத்தப்படுகின்றன, இதனால் அமைப்பு முக்கியமான தரவை அடையாளம் காணவும், தனித்துவமான ஆவண கைகேசைகளை உருவாக்கவும், பாரம்பரிய அமைப்புகள் தவறவிடக்கூடிய அங்கீகரிக்கப்படாத மாற்றங்கள் அல்லது உள்ளடக்கப் பிரித்தெடுத்தலைக் கண்டறியவும் உதவுகிறது. பாதுகாப்பை மேலும் வலுப்படுத்த, சேமிப்பு மற்றும் பரிமாற்றத்தின் போது ஆவணங்கள் பாதுகாப்பாக இருப்பதை உறுதிசெய்ய, அமைப்பு எலிப்டிகல் வளைவு குறியாக்கத்தை ஒருங்கிணைக்கிறது, இதனால் அங்கீகரிக்கப்படாத பயனர்களால் அவற்றைப் படிக்க முடியாது. கூடுதலாக, இந்தத் திட்டத்தில் யதார்த்தமான போலி ஆவணங்களை உருவாக்கி, சாத்தியமான தாக்குபவர்களை தவறாக வழிநடத்த மாற்று களஞ்சியங்களில் சேமிப்பதன் மூலம் ஒரு ஏமாற்று உத்தி அடங்கும். உள்ளடக்க அடிப்படையிலான கண்காணிப்பு, வலுவான குறியாக்கம் மற்றும் ஏமாற்று வரிசைப்படுத்தல் ஆகியவற்றை இணைக்கும் இந்த கலப்பின அணுகுமுறை முக்கியமான தகவல்களைப் பாதுகாப்பதற்கான விரிவான மற்றும் முன்முயற்சியுடன் கூடிய தீர்வை உருவாக்குகிறது. ஆவணத் திருட்டு மற்றும் அங்கீகரிக்கப்படாத மாற்றங்களை எதிர்ப்பதில் இது குறிப்பிடத்தக்க முன்னேற்றத்தை வழங்குகிறது, இது தரவு ரகசியத்தன்மை மற்றும் ஒருமைப்பாட்டிற்கு முன்னுரிமை அளிக்கும் தொழில்களில் மிகவும் பொருந்தக்கூடியதாக அமைகிறது.

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LIST OF ABBREVIATIONS

S. NO	ABBREVIATION	EXPLANATION
1	NLP	Natural Language Processing
2	ECC	Elliptic Curve Cryptography
3	TF	Term Frequency
4	IDF	Inverse Document Frequency
5	IDS	Intrusion Detection Systems
6	VPN	Virtual Private Network
7	RSA	Rivest–Shamir–Adleman
8	IP	Intellectual Property
9	DOS	Denial-Of-Service
10	MITM	Man-In-The-Middle
11	BAW	Bayesian Active Watermarking
12	MTD	Moving Target Defense
13	NER	Named Entity Recognition
14	MFA	Multi- Factor Authentication
15	UML	Unified Modeling Language
16	DFD	Data Flow Diagram

CHAPTER 1

INTRODUCTION

In the modern digital landscape, protecting sensitive documents from theft, unauthorized access, and tampering is a major concern across various sectors. Traditional document security measures such as passwords and basic encryption often fall short when faced with sophisticated cyber threats. As the volume and importance of digital documents grow, there is an increasing need for a comprehensive security system that goes beyond access control and static protection mechanisms. This project, DARD – Defense Against IP Theft, addresses these challenges by introducing a multi-layered strategy to safeguard critical information. The proposed solution integrates Natural Language Processing (NLP) and cryptographic techniques to ensure both proactive and reactive defense mechanisms.

NLP plays a key role in analyzing document contents using TF-IDF (Term Frequency–Inverse Document Frequency), helping identify sensitive keywords and understand contextual relevance. This enables intelligent monitoring, classification, and alerting of potential risks related to unauthorized modifications or content extraction. Simultaneously, the system creates realistic decoy documents to deceive attackers and divert them from accessing actual sensitive files, adding an extra layer of protection. To further strengthen the system, Elliptical curve cryptography encryption is used to secure the storage and transmission of real documents, making them accessible only to authorized users. This combination of content analysis, encryption, and deception creates a holistic defence mechanism that is adaptive, secure, and efficient. By implementing this model, organizations can better manage and protect their digital documents, reduce the chances of data breaches, and ensure the integrity and confidentiality of valuable information.

Network Security

Network security is a crucial aspect of modern digital communication systems, aimed at protecting the integrity, confidentiality, and availability of data as it travels across or is stored within computer networks. With the rapid growth of internet-based services, cloud computing, and wireless technologies, the risk of cyber threats has significantly increased. Network security encompasses a range of technologies, tools, and processes designed to defend against unauthorized access, misuse, modification, or destruction of data and systems. A secure network ensures that only authorized users and devices can access the network and its resources. This involves the use of firewalls, intrusion detection systems (IDS), virtual private networks (VPNs), and encryption techniques to protect sensitive information from being intercepted or tampered with. Network security also includes policies and practices for monitoring network traffic, detecting vulnerabilities, and responding to potential threats promptly to minimize damage.

Cyberattacks such as phishing, malware, ransomware, denial-of-service (DoS) attacks, and man-in-the-middle (MITM) attacks have become more sophisticated, targeting individuals, businesses, and even governments. As a result, the demand for robust and proactive network security measures has grown. Organizations must implement layered security models, also known as defense-in-depth, to provide multiple lines of defense against potential intrusions or breaches. In addition to technological solutions, network security also requires user education, regular software updates, and compliance with industry regulations and standards. Effective network security not only protects data but also builds trust among users, clients, and stakeholders by ensuring that sensitive information is safeguarded against evolving cyber threats. As the digital landscape continues to evolve, so must network security strategies to keep pace with emerging technologies and threats.

Natural Language Processing (NLP)

Natural Language Processing (NLP) plays a central role in enhancing document security by enabling intelligent, content-aware analysis and monitoring. In the proposed system, NLP techniques such as TF-IDF (Term Frequency–Inverse Document Frequency) are employed to scan and understand the content of documents in real time. This allows the system to identify sensitive keywords, phrases, and contextual patterns that could indicate critical or confidential information. By quantifying the importance of specific terms relative to a corpus of documents, TF-IDF helps in flagging content that may need additional protection. This enables proactive classification and tagging of documents based on their sensitivity level, rather than relying solely on static metadata or user-defined labels. Moreover, the NLP engine can detect unusual textual patterns or modifications that may signal an attempt at tampering or unauthorized content extraction. Beyond content classification, NLP also supports the generation of unique document fingerprints essentially semantic signatures that allow the system to track changes, verify integrity, and detect plagiarism or data leakage.

By continuously monitoring document interactions and comparing live content against previously stored fingerprints, the system can issue alerts when discrepancies are found. This real-time contextual analysis surpasses traditional security systems that only control access or encrypt content without understanding its meaning. Additionally, NLP is employed in crafting decoy documents that are contextually realistic. These fake documents are generated using linguistic patterns similar to genuine sensitive files, misleading attackers and increasing the likelihood of engagement with non-critical data. Thus, NLP not only strengthens security through intelligent content analysis but also enhances deception strategies, making the overall defense system robust, adaptive, and highly efficient.

Elliptic Curve Cryptography (ECC)

Elliptic Curve Cryptography (ECC) is a form of public-key cryptography that offers high security with relatively smaller key sizes compared to traditional cryptographic methods such as RSA. The core principle of ECC lies in the mathematical properties of elliptic curves over finite fields, which provide a more efficient and robust method for securing digital communications. ECC enables secure encryption, digital signatures, and key exchange mechanisms with shorter key lengths, making it computationally faster and more energy-efficient, especially in environments with limited resources like mobile devices or IoT systems. The small key sizes in ECC do not compromise security, as they offer the same level of protection as longer RSA keys, making ECC a preferred choice for modern cryptographic applications.

In the context of document security, ECC is utilized to enhance the encryption of sensitive data during storage and transmission. By using elliptic curve-based algorithms, documents are encrypted in a way that makes them unreadable to unauthorized users. The ECC keys allow for secure data exchange without the need for heavy computational resources, which is crucial in scenarios where large volumes of sensitive information need to be transmitted securely. ECC also facilitates the generation of digital signatures, which ensure both the authenticity and integrity of documents. By employing ECC for document protection, organizations can achieve a high level of security while maintaining efficiency and reducing the processing overhead associated with other encryption methods. This makes ECC an ideal solution for safeguarding intellectual property, ensuring that only authorized users can access or modify critical documents.

Primary Focus

The primary focus of the project is to create a comprehensive and intelligent system for securing sensitive digital documents against unauthorized access, tampering, and intellectual property theft. In the face of increasingly sophisticated cyber threats, traditional defenses such as basic encryption and access control are proving inadequate. To address this gap, the project introduces a multi-layered defense mechanism that integrates Natural Language Processing (NLP), Elliptic Curve Cryptography (ECC), and deception strategies involving the generation of fake documents. This combination allows for the real-time analysis of document content, proactive identification of sensitive information, strong encryption for secure storage and transmission, and the use of realistic decoys to mislead potential attackers.

Extent Of Work

The project includes the development and implementation of a secure document handling system capable of classifying, encrypting, tracking, and managing documents within an organization. It encompasses not only content-based sensitivity detection and strong cryptographic protection but also real-time monitoring of user interactions, alerting mechanisms for suspicious activities, and comprehensive access control through multi-factor authentication. The system supports administrative and user interfaces for uploading, encrypting, and retrieving documents while maintaining rigorous auditing and notification procedures. Its applications are relevant across various industries where document integrity and confidentiality are paramount, such as legal, healthcare, corporate R&D, and government sectors.

Target Outcomes

The project ensure that critical digital documents remain secure throughout their lifecycle by implementing intelligent monitoring, robust encryption, and deception techniques. The system is designed to identify sensitive content dynamically using NLP techniques like TF-IDF, enforce access restrictions through cryptographic authentication, and detect any unauthorized modifications or extraction attempts through document fingerprinting and integrity checks. By combining these elements, the project aims to enhance organizational resilience against cyber intrusions and establish a reliable framework for digital document protection.

Additionally, the objective of this project extends to creating a proactive defense infrastructure that not only reacts to threats but anticipates and neutralizes them before any damage occurs. By leveraging NLP for semantic understanding and ECC for secure communication, the system moves beyond conventional security models to offer intelligent, context-aware protection. It also seeks to empower administrators with actionable insights by tracking document interactions, detecting anomalies, and maintaining detailed logs for auditing and compliance purposes. The integration of decoy documents further adds a strategic layer of deception, diverting malicious intent away from real data and providing early warning signals of potential breaches. Ultimately, the objective is to build a resilient, scalable, and adaptable security solution that evolves with emerging threats and organizational needs.

CHAPTER 2

LITERATURE SURVEY

2.1 "DARD: Deceptive Approaches for Robust Defense Against IP Theft (2024) " – Alberto Maria Mongardini, Et Al.

The paper introduces DARD, a defensive framework against intellectual property (IP) theft that deceives adversaries using automated document analysis. Attackers typically exfiltrate large volumes of sensitive data and apply machine learning tools like topic modelling and clustering to extract valuable content. DARD counteracts this by generating a deceptive repository using four operations: Basic Shuffle, Shuffle Increment, Shuffle Reduction, and Change Topic. These techniques alter keyword structures and document groupings, confusing tools like Amazon Comprehend and Latent Dirichlet Allocation. As a result, adversaries receive distorted clusters that hinder topic identification and require costly manual review. Legitimate users access original content through a secure enclave that maps deceptive keywords back to the true data, maintaining usability. Even if compromised, the deceptive repository renders attacker results unreliable. Experimental validation shows DARD successfully disrupts multiple attack models. It proves effective across commercial and open-source analysis tools. The system reduces the accuracy of automated data extraction and delays attacker progress. DARD emphasizes data-centric deception rather than relying solely on perimeter defenses. It's a low-cost, proactive solution that protects corporate IP without disrupting authorized use. This novel approach enhances digital security by increasing adversarial workload and reducing their success. DARD is a cost-effective, proactive solution that shifts the burden to attackers. It reinforces the security of valuable digital assets without compromising legitimate functionality. Overall, it offers a novel, practical framework for defending sensitive corporate knowledge from automated exploitation.

2.2 "The Role of Cybersecurity in Protecting Intellectual Property (2024) "

– Chirag Mavani, Et Al.

This paper explores the critical role of cybersecurity in safeguarding intellectual property (IP) in the digital era. As businesses increasingly rely on digital infrastructures, the protection of proprietary assets like patents, trade secrets, and source codes becomes paramount. The study highlights various cybersecurity threats that endanger IP, including cyber espionage, insider threats, and ransomware attacks. The authors review current defense mechanisms and emphasize the need for layered security frameworks, incorporating firewalls, intrusion detection systems, encryption, and access controls. The paper presents real-world case studies where cybersecurity lapses led to IP theft, reinforcing the importance of proactive defense strategies. It also evaluates the impact of government policies and legal frameworks on IP protection. The researchers recommend integrating cybersecurity awareness training into corporate culture. Moreover, the study outlines how emerging technologies like AI and blockchain can bolster IP protection by enabling real-time threat detection and secure auditing. A key contribution is the proposed cybersecurity-IP integration model, which aligns IT security protocols with IP asset classification. The paper underscores the need for ongoing vulnerability assessments and robust incident response plans.

2.3 "A Psycholinguistics-Inspired Method to Counter IP Theft Using Fake Documents (2024) "– Natalia Denisenko, Et Al.

This innovative study presents a novel psycholinguistics-based approach to intellectual property theft mitigation by strategically deploying fake documents. The method is grounded in cognitive deception and linguistic manipulation to mislead adversaries attempting to steal proprietary data. These decoy documents are crafted using psycholinguistic principles to appear

authentic, increasing the likelihood of interception while leaving real IP untouched. The authors detail an algorithm that dynamically generates fake documents embedded with behavioral triggers to alert organizations when accessed. The study presents a threat model of adversaries engaging in document exfiltration and how psychological misdirection through linguistic mimicry can delay or divert them. Realistic evaluation scenarios demonstrate the effectiveness of the method in both protecting real assets and enabling intrusion detection. A highlight is the system's adaptability—fake documents evolve with the context of the organization's operations. The authors also address ethical implications, noting how the technique aligns with deterrence strategies in cybersecurity. This work contributes a fresh interdisciplinary angle, merging language science with cybersecurity practices. It calls for more research into deception technologies and their role in modern data protection. Experimental results confirm increased time-to-theft and confusion among attackers. This approach also reduces reliance on perimeter security.

2.4 "Reliable Model Watermarking: Defending Against Theft Without Compromising on Evasion (2024) " – Hongyu Zhu, Et Al.

This paper introduces a reliable watermarking technique for protecting machine learning models against intellectual property theft, focusing on maintaining model integrity while preserving its evasion capabilities. The authors develop a watermarking strategy that embeds imperceptible triggers within the model's decision logic. These triggers are activated under controlled conditions to verify ownership without affecting model predictions on benign inputs. The approach strikes a critical balance—providing verifiability without degrading performance or making the model susceptible to detection. A key innovation is the integration of a dual-objective training paradigm that optimizes for both accuracy and watermark robustness. The study evaluates the method across

several deep learning models and attack scenarios, including fine-tuning, pruning, and knowledge distillation. Results show that the proposed watermarking technique withstands various model extraction and evasion attacks. The paper also addresses legal implications by ensuring watermarks are admissible for IP claims. By designing watermarks that can be verified without public disclosure, the system promotes secure ownership claims. The authors emphasize minimal computational overhead and compatibility with existing model architectures. Their watermarking protocol is modular and easy to integrate with standard training pipelines.

2.5 "EMMARK: Robust Watermarks for IP Protection of Embedded Quantized Large Language Models (2024)" – Ruisi Zhang, Et Al.

EmMark is presented as a novel solution for embedding robust watermarks in quantized large language models (LLMs), especially designed for deployment on embedded systems. The paper addresses the unique challenge of watermarking LLMs post-quantization—a process that often degrades the fidelity of standard watermarking techniques. The authors propose a model-aware watermarking technique that adapts to the quantization behaviour of LLMs, ensuring watermark survivability even in resource-constrained environments. EmMark supports ownership verification, tracing, and tamper detection without affecting model accuracy. The methodology involves injecting functional triggers that generate specific outputs when queried with secret keys. These triggers are optimized through reinforcement learning to ensure resilience against fine-tuning and compression. The authors conduct comprehensive experiments using models like GPT-based architectures and validate the effectiveness under multiple attack vectors. Their results demonstrate over 90% trigger response consistency across multiple deployment settings. The paper further explores the integration of EmMark in secure hardware platforms for embedded applications. Additionally,

the watermarking strategy is shown to be undetectable through traditional adversarial inspection methods. This approach bridges a crucial gap in IP protection of LLMs used in commercial edge-AI applications.

2.6 "Defense Against Model Extraction Attack by Bayesian Active Watermarking (2024) " – Zhenyi Wang, Et Al.

This research introduces a Bayesian Active Watermarking (BAW) approach aimed at countering model extraction attacks—a rising threat to AI intellectual property. BAW utilizes Bayesian principles to embed watermarks that adaptively trigger when queried by suspicious inputs typical of extraction attempts. The approach combines active learning and uncertainty estimation to maximize detection efficiency while minimizing the impact on normal queries. The paper explains how BAW selectively chooses watermark triggers based on model confidence scores and adversarial behaviour patterns. These triggers function as legal ownership proofs and intrusion alerts, helping model owners identify unauthorized usage. The method is tested across classification and regression models, including convolutional neural networks and gradient-boosted trees. Evaluation metrics include watermark fidelity, evasion robustness, and false-positive rates. Results show significant performance in identifying extraction without degrading model outputs. The authors also examine how BAW adapts to changing adversarial strategies over time. Importantly, the system operates with negligible overhead, making it suitable for real-world deployment. The inclusion of dynamic trigger rotation enhances security further. This work is a leap toward practical and intelligent IP protection for commercial AI models. The approach aims to strike a balance between preserving model performance for legitimate users while defending against adversarial replication, thus enhancing the security and intellectual property protection of deployed AI systems.

2.7 "PASSP: A Private Authorization Scheme Oriented Service Providers (2024) " – Weiqi Dai, Et Al.

The document proposes a decentralized authorization scheme designed to enhance user privacy in data-sharing ecosystems. Unlike traditional systems relying on centralized intermediaries, PASSP uses a blockchain-based framework with publicly verifiable proxy re-encryption and IPFS for decentralized storage. This enables users to authorize data access while concealing their identity and intent. Zero-knowledge proofs (ZKPs) verify authorization without revealing private details. Designed for trustless environments and aligned with Web 3.0 principles, PASSP gives users full control over their data access with transparent, on-chain verifiable records for public auditing. It prevents service providers from accessing sensitive metadata and supports both one-time and multiple-use access. A full Ethereum implementation using zk-SNARKs demonstrates the scheme's security and practicality, with low computational and communication overhead. Blockchain integration ensures tamper-proof auditing and incentivizes correct behavior. PASSP advances privacy by separating data control from service logic, offering scalable and efficient privacy-preserving authorization without centralized oversight.

2.8 "Privacy-Preserving Multi-Biometric Indexing Based on Frequent Binary Patterns (2024) " – Dailé Osorio-Roig, Et Al.

The document introduces an efficient and scalable system addressing privacy and computational challenges in multi-biometric identification. The system uses frequent binary patterns (FBPs) to index cancelable biometric templates, significantly reducing search space and comparison operations compared to traditional exhaustive searches. It supports multiple biometric traits like face, iris, and fingerprint, and is agnostic to template protection schemes. By applying feature- and representation-level fusion, it creates robust indexes

combining different biometrics. Cancelable templates ensure irreversibility and unlinkability, meeting privacy standards such as GDPR. The approach reduces computational workload by up to 57% with three modalities and 53% with two, without sacrificing accuracy; biometric performance improves at high-security levels. Evaluations align with ISO standards, using DNN-based embeddings and binary transformations to retain high discrimination. FBPs enable efficient binning and accelerate identification without decrypting data, maintaining security. The system supports single-step identification across modalities and offers a practical, secure solution for large-scale biometric identification, balancing privacy, performance, and efficiency.

2.9 "DOLOS: A Novel Architecture for Moving Target Defense (2023) " – Giulio Pagnotta, Et Al.

This extended work revisits Dolos, an adaptive Moving Target Defense (MTD) architecture designed to dynamically shift the attack surface of computing systems. It reaffirms the key principle of constantly reconfiguring system attributes such as IP addresses, port numbers, and service endpoints to confuse and delay adversaries. The paper presents a more mature version of Dolos with enhanced orchestration mechanisms using containerization tools like Docker and Kubernetes, allowing seamless re-deployment and migration of services. The defense strategy is built on a layered security approach incorporating monitoring, anomaly detection, and reactive mitigation, ensuring that threats are not only deflected but also logged and analyzed. An in-depth performance evaluation shows that Dolos imposes minimal latency while significantly increasing attacker workload and reconnaissance time. With support for policy-driven automation, Dolos adapts to risk levels in real-time, making it highly suited for environments requiring resilience and operational continuity, such as critical infrastructure and cloud-hosted services.

2.10 "TRUSTEE: Towards the Creation of Secure, Trustworthy and Privacy-Preserving Framework (2023) " – Sarwar Sayeed, Et Al.

TRUSTEE is revisited as a unified cybersecurity framework tailored for trust, privacy, and data integrity in distributed digital ecosystems. The paper builds on a modular architecture integrating key components like attribute-based access control (ABAC), blockchain-based audit trails, and homomorphic encryption to allow computations over encrypted data. TRUSTEE's design is guided by regulatory frameworks like GDPR, ensuring legally compliant data handling across sectors including healthcare, IoT, and public services. The authors introduce a federated identity management system that enables decentralized authentication while preserving user anonymity. TRUSTEE's privacy-preserving data sharing protocols and real-time threat detection components help organizations respond to data misuse and internal threats effectively. A comprehensive case study in a multi-agency smart governance system illustrates the flexibility and robustness of the framework. The study demonstrates enhanced traceability, reduced insider risk, and improved user trust, positioning TRUSTEE as a foundational architecture for secure digital transformation.

2.11 "Toward Effective Evaluation of Cyber Defense: Threat-Based Adversary Emulation Approach (2023) " – Abdul Basit Ajmal, Et Al.

This paper proposes a cyber defense evaluation methodology that leverages threat-based adversary emulation as a reliable metric for assessing organizational security postures. The framework is grounded in the ATT&CK framework by MITRE, allowing realistic simulation of attacker behaviors like lateral movement, privilege escalation, and data exfiltration. The emulation approach bridges the gap between traditional red teaming and automated security

validation tools, offering repeatable, data-driven testing. The proposed model includes attack orchestration engines, telemetry collection tools, and reporting dashboards to visualize defensive gaps. The study evaluates its methodology on enterprise-grade systems and shows how even well-secured environments exhibit blind spots in detection pipelines. By emphasizing threat-informed defense, the approach helps organizations prioritize investments in cybersecurity and tailor their detection mechanisms to match evolving threat vectors. The research highlights how adversary emulation facilitates continuous security improvement by testing detection logic under controlled and reproducible scenarios.

2.12 "A Comprehensive Model for Securing Sensitive Patient Data in a Clinical Scenario (2023) " – Antonio López Martínez, Et Al.

Focusing on healthcare cybersecurity, this paper presents a multi-layered security model to protect sensitive patient data throughout its lifecycle—from collection to transmission and storage. The model utilizes encryption, anonymization, and context-aware access control to secure data in compliance with privacy regulations like GDPR and HIPAA. One of its novel features is the real-time access validation mechanism, which uses contextual information such as clinician role, urgency, and patient consent to decide access rights. A secure audit mechanism logs all interactions with data, supporting legal accountability and forensic investigation. The architecture supports interoperability with existing EHR systems using HL7 and FHIR standards and offers a flexible deployment model for cloud or on-premise use. The paper includes a comparative performance analysis showing negligible latency increase and high access precision. Overall, this security model enhances trustworthiness and compliance in clinical workflows, offering a template for privacy-first digital health ecosystems.

2.13 "Is Your Model Sensitive? SPEDAC: A New Resource for the Automatic Classification of Sensitive Personal Data (2023) " – Gaia Gambarelli, Et Al.

This paper introduces SPEDAC, a pioneering benchmark dataset for the automatic classification of Sensitive Personal Data (SPD) in natural language. It addresses the growing need to align machine learning models with privacy standards by identifying and managing data that could expose personal information. SPEDAC includes diverse text samples from legal, medical, governmental, and social contexts, annotated for sensitivity categories such as racial/ethnic origin, health data, political opinions, and union membership. The authors evaluate various deep learning models including transformers (BERT, RoBERTa) and logistic regression baselines, analyzing their ability to detect SPD in free-form text. The study finds that contextual embeddings outperform traditional models, particularly in ambiguous or domain-specific contexts. Beyond classification accuracy, the paper also explores model fairness and explainability, emphasizing the ethical responsibility in automated data processing. SPEDAC sets a foundation for privacy-aware NLP pipelines and can support applications in data anonymization, compliance auditing, and AI governance.

CHAPTER 3

SYSTEM STUDY

3.1 Existing System

Traditional document security systems primarily rely on reactive defense mechanisms such as firewalls, intrusion detection systems (IDS), and basic encryption protocols. These methods are often focused on protecting network perimeters or restricting unauthorized access through passwords and user permissions. However, with the evolving nature of cyber threats, these approaches are increasingly proving insufficient. Attackers today are more sophisticated, utilizing advanced tactics to bypass access controls and infiltrate sensitive document repositories. The lack of active monitoring and contextual content analysis in these systems means that breaches often go undetected until the damage is already done.

An emerging solution in the cybersecurity landscape is the concept of deceptive repositories, which aim to proactively mislead and trap attackers by creating decoy assets. These repositories are filled with fake but realistic-looking documents, databases, and credentials designed to mimic genuine data. When attackers interact with these decoys, alerts are triggered, enabling cybersecurity teams to monitor and analyze the behavior of potential threats in a controlled environment. One such tool, WE-Forge, employs word embeddings to generate convincing fake documents by preserving semantic and syntactic relationships between words. This ensures that the decoy content is contextually relevant to the targeted domain, effectively confusing and slowing down attackers. Despite their innovative approach, deceptive repository systems have notable limitations. For instance, in highly specialized fields, generating fake documents that convincingly mimic real ones can be challenging without in-depth domain knowledge. Additionally, if the training corpus for tools like WE-Forge is

outdated or not representative of the actual domain, the realism of the decoy documents suffers, reducing their effectiveness. There's also the risk that legitimate users may mistakenly interact with decoy files, leading to confusion or operational disruption. Thus, while deception-based defenses offer a new layer of security, they must be carefully maintained and supplemented with more robust content analysis and encryption mechanisms to ensure complete and reliable document protection.

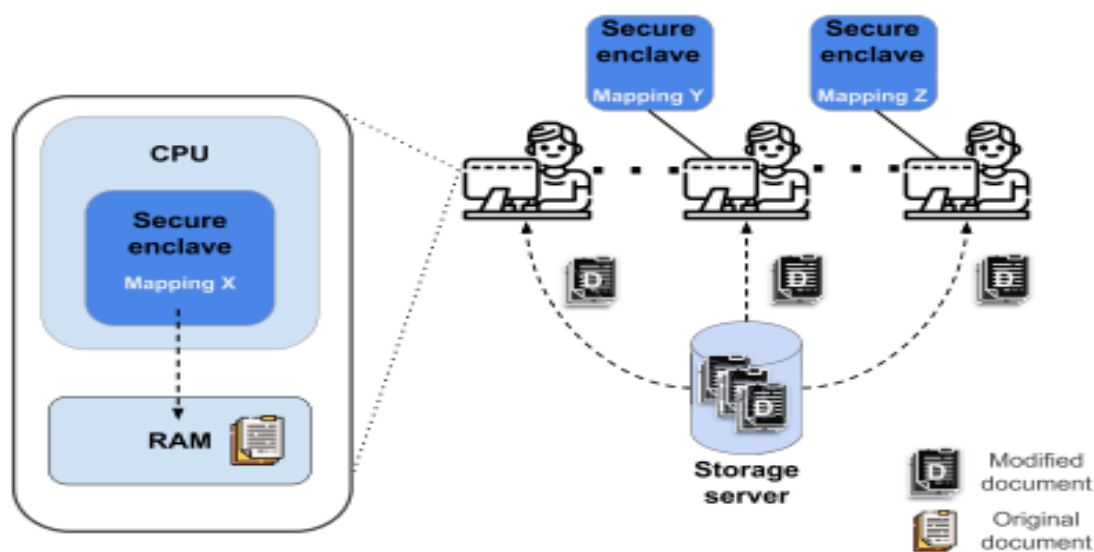


Figure 3.1.1: Secure Document Processing Using Enclave-Based Isolation and Mapping

Disadvantages

- Relies heavily on traditional defenses like firewalls and passwords, which are easy to bypass.
- Lacks real-time monitoring and contextual analysis of document content.
- Fake document generation may fail in specialized domains without domain-specific knowledge.
- Outdated training data can reduce the realism and effectiveness of decoy files.

3.2 Proposed System

The proposed system introduces a dual-layered defense mechanism by integrating Natural Language Processing (NLP) and cryptographic techniques to ensure robust document security. NLP is used to analyze document content using the TF-IDF (Term Frequency-Inverse Document Frequency) algorithm, enabling the system to identify and classify sensitive data based on keyword relevance and contextual significance. This intelligent analysis allows real-time monitoring of user interactions with documents, helping to detect any unauthorized attempts to extract, modify, or misuse critical information. By understanding the content and its importance, the system ensures that high-priority documents are given greater protection.

In addition to content analysis, the system incorporates deception techniques by generating fake documents and storing them in alternative repositories. These decoy documents mimic the structure and context of genuine documents but contain no real sensitive data. The purpose of this deception layer is to mislead unauthorized users or potential attackers, causing them to engage with false information instead of actual intellectual property. This not only protects the genuine assets but also allows cybersecurity teams to track intrusion attempts and gather insights into attacker behavior. To further fortify the security, ECC encryption is applied to all sensitive documents. ECC is a symmetric-key block cipher known for its high speed and strong security, ensuring that only users with the correct decryption key can access the protected data. This encryption layer guarantees the confidentiality and integrity of documents during both storage and transmission. By combining NLP for intelligent content monitoring, deception tactics for misdirection, and ECC encryption for secure access control, the proposed system delivers a comprehensive and proactive solution for defending against document theft and unauthorized access.

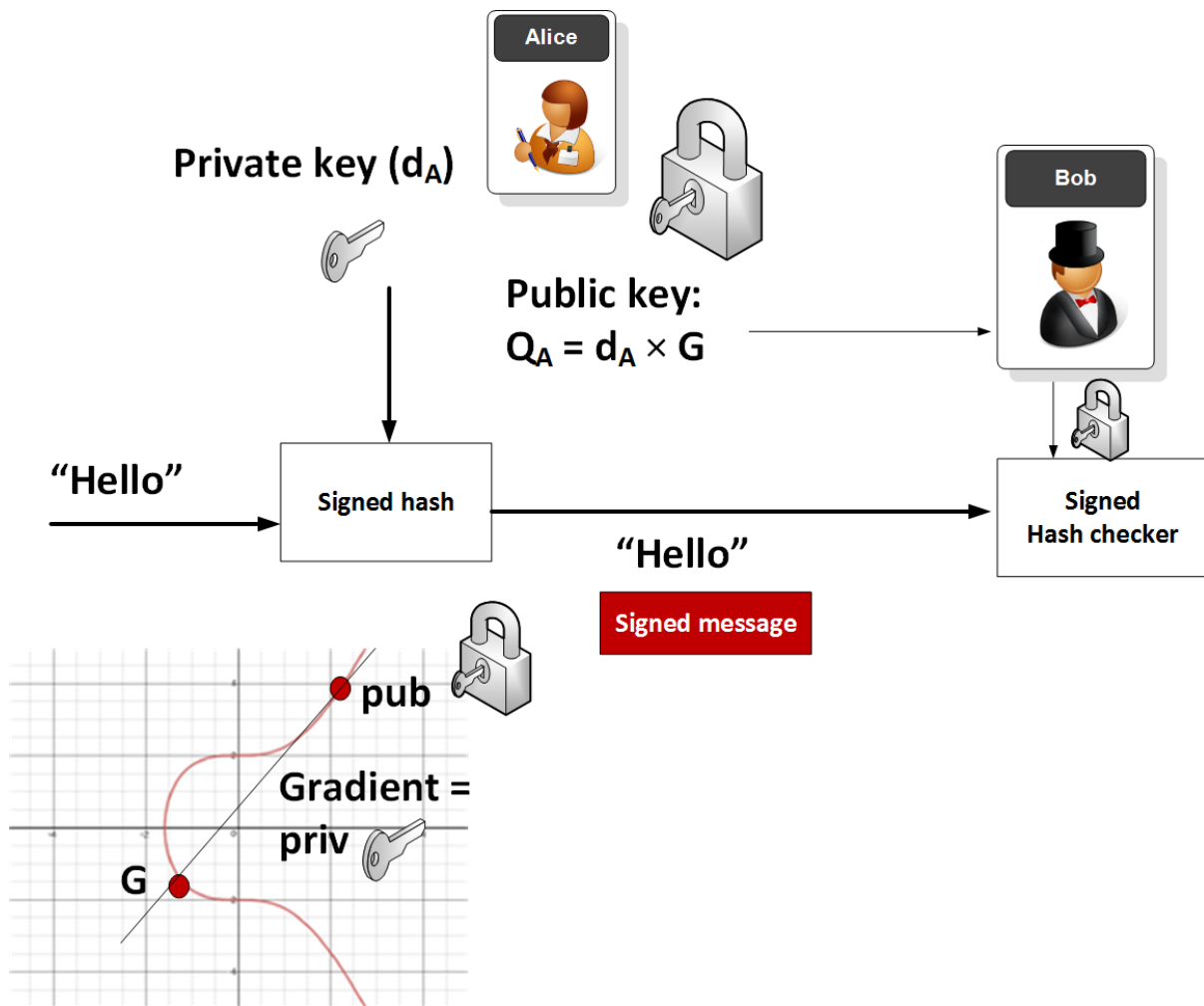


Figure 3.2.1: ECC Encryption Process

Advantages

- Provides a multi-layered defence using NLP, deception, and encryption.
- Detects unauthorized access through real-time content monitoring.
- Misleads attackers with decoy documents stored in alternative locations.
- Secures sensitive files using strong ECC encryption.
- Enhances organizational resilience against data breaches and theft.

CHAPTER 4

SYSTEM SPECIFICATIONS

4.1 Hardware Requirements

- Processor : Intel processor 2.6.0 GHZ
- RAM : 4 GB
- Hard disk : 160 GB
- Keyboard : Standard keyboard
- Monitor : 15- inch color monitor

4.2 Software Requirements

- Server Side : Python 3.7.4(64-bit) or (32-bit)
- Client Side : HTML, CSS, Bootstrap
- IDE : PyCharm
- Back end : MySQL 5.
- Server : WampServer 2i
- OS : Windows 10 64 –bit

HTML

Hypertext Markup Language (HTML) is the standard markup language for documents designed to be displayed in a web browser. It can be assisted by technologies such as Cascading Style Sheets (CSS) and scripting languages such as JavaScript. Inclusion of CSS defines the look and layout of content. HTML provides a means to create structured documents by denoting structural semantics for text such as headings, paragraphs, lists, links, quotes and other items.

CSS

CSS, or Cascading Style Sheets, is a language used for describing the presentation of a document written in HTML (Hyper Text Markup Language). It controls how HTML elements are displayed on a webpage, including their layout, colors, fonts, and more. CSS allows you to separate the structure of a webpage from its design, making it easier to maintain and update. You can apply CSS rules directly within an HTML document, or link to an external CSS file for greater reusability and organization. CSS uses selectors to target specific HTML elements and apply styling rules to them. These rules consist of properties and values, defining aspects such as size, color, spacing, and positioning. CSS also supports various techniques like media queries for creating responsive designs that adapt to different screen sizes and devices.

Bootstrap

Bootstrap is a free and open-source tool collection for creating responsive websites and web applications. It is the most popular HTML, CSS, and JavaScript framework for developing responsive, mobile-first websites. It solves many problems which we had once, one of which is the cross-browser compatibility issue. Nowadays, the websites are perfect for all the browsers (IE, Firefox, and

Chrome) and for all sizes of screens (Desktop, Tablets, Phablets, and Phones). All thanks to Bootstrap developers -Mark Otto and Jacob Thornton of Twitter, though it was later declared to be an open-source project.

Python Flask

Flask is a web framework. This means flask provides you with tools, libraries and technologies that allow you to build a web application. This web application can be some web pages, a blog, a wiki or go as big as a web-based calendar application or Flask is-often referred to as a micro framework. It aims to keep the core of an application simple yet extensible. Flask does not have built-in abstraction layer for database handling, nor does it have formed a validation support. Instead, Flask supports the extensions to add such functionality to the application. Although Flask is rather young compared to most Python frameworks, it holds a great promise and has already gained popularity among Python web developers. Let's take a closer look into Flask, so-called "micro" framework for Python. Flask was designed to be easy to use and extend. The idea behind Flask is to build a solid foundation for web applications of different complexity. From then on you are free to plug in any extensions you think you need. Also you are free to build your own modules. Flask is great for all kinds of projects. It's especially good for prototyping.

MySQL

MySQL is an open-source relational database management system that uses SQL (Structured Query Language) to store, manage, and retrieve data. It organizes information into tables consisting of rows and columns, allowing for efficient data handling and complex queries. MySQL supports relationships between tables using keys, which helps maintain data integrity. Known for its speed, reliability, and ease of use, it has become a popular choice for developers.

It powers many web applications, including well-known platforms like WordPress and Facebook. MySQL runs on various operating systems such as Windows, Linux, and macOS. It is capable of handling large volumes of data and supports features like transactions, replication, and security. Originally developed by MySQL AB, it is now owned and maintained by Oracle Corporation.

PyCharm

PyCharm is a powerful IDE developed by JetBrains for Python programming. It offers intelligent code completion, debugging, and testing tools to streamline development. With support for web frameworks like Django, it helps build web apps efficiently. PyCharm also integrates with version control systems like Git. It is available in both free (Community) and paid (Professional) editions. The IDE also supports database tools and web technologies in the Professional version. Its user-friendly interface and productivity features make it a favorite among Python developers.

WampServer

WampServer is a Windows-based software stack that lets developers create web applications using Apache, MySQL, and PHP. It provides an easy-to-use interface to manage services and configure server settings. WampServer is ideal for testing and developing websites locally before deploying them. It includes phpMyAdmin for database management through a browser. The installation is simple, making it suitable for beginners and professionals alike. It allows developers to switch between different versions of PHP, MySQL, and Apache. WampServer is a convenient tool for creating and managing a full web development environment on a local machine.

CHAPTER 5

MODULES DESCRIPTION

5.1 List of Modules

5.1.1 User Interface Module

5.1.2 Classification Module

5.1.3 Encryption and Authentication Module

5.1.4 Access Control and Audit Module

5.1.5 Document Tracking and Integrity Monitoring Module

5.1.6 Alert and Notification Module

5.2 Modules Description

5.2.1 User Interface Module

This module provides a user-friendly interface for both administrators and authorized users to interact with the system efficiently. It allows users to upload their Intellectual Property (IP) documents directly through the platform. Once uploaded, users can encrypt documents and manage their metadata securely. Administrators can monitor document activity and real-time security events. The interface supports the customization of access permissions based on user roles. Users can also set their preferences, including notification types, language, and interface themes. The design emphasizes simplicity while ensuring full functionality. Navigation is intuitive, making document handling seamless. This module acts as the control hub for all user-level operations.

5.2.2 Classification Module

The Document Content Analysis module uses advanced Natural Language Processing (NLP) techniques to examine and classify content based on its sensitivity. It identifies proprietary, financial, or personal data within the

uploaded document. Named Entity Recognition (NER) is used to highlight sensitive fields like names, account numbers, and locations. The module generates a unique fingerprint for every document by analyzing its structure, keywords, and patterns. This fingerprint is securely stored for future verification and integrity checks. It continuously scans for unauthorized modifications to the document content. Any suspicious change triggers alerts in real-time. The goal is to ensure that data within the document remains intact and confidential. This module enhances document intelligence and traceability.

5.2.3 Encryption and Authentication Module

This module ensures data confidentiality by applying ECC encryption to all uploaded documents before storage or transmission. ECC provides strong security using smaller keys, making it faster and efficient compared to other encryption methods. The encrypted files are difficult to decrypt without the proper key, ensuring unauthorized users are blocked. Digital signatures are added to verify the authenticity of the sender and detect tampering. This module also manages the decryption process using the recipient's private key, enabling only authorized users to view the content. It protects the document during both transit and rest. Security policies are enforced consistently. Encryption and decryption are seamless to users. This layer is crucial for IP protection.

5.2.4 Access Control and Audit Module

The Access Control and User Authentication module safeguards the system from unauthorized access by implementing multi-factor authentication (MFA). Biometric, password, and private key-based methods can be used based on system settings. It defines roles and permissions clearly, so different users have different levels of access. For example, only an admin may edit while others can view. It tracks every access, login, and document modification with accurate timestamps. Logs are stored for auditing and forensic analysis. This module

ensures accountability for all actions within the system. It restricts sensitive operations to specific roles. Tampering or unusual access patterns are flagged instantly. The module helps prevent insider threats and breaches.

5.2.5 Document Tracking and Integrity Monitoring Module

This module tracks the lifecycle of each document, including when and where it is accessed, edited, or shared. It captures metadata such as IP address, timestamp, and user credentials for every activity. Integrity verification is done by comparing the current state of the document with its previously stored fingerprint. If any deviation is found, it implies possible tampering, and an alert is generated. This ensures that even subtle unauthorized changes are not missed. Document flow across systems is monitored in real-time. It supports audit trails for legal and compliance requirements. It helps in quickly identifying the point of breach. Tracking is continuous and automated. This module maintains transparency and trust.

5.2.6 Alert and Notification Module

The Alert and Notification System provides real-time updates regarding document activity and potential threats. It sends notifications when unauthorized users attempt to access or alter any document. Alerts are also triggered by suspicious content extraction or structural changes in a document. The system can be configured to notify through emails, dashboard pop-ups, or SMS. Periodic reports are sent to administrators and users summarizing document security events. It ensures immediate awareness of ongoing breaches. Notification settings can be customized based on user roles and preferences. The system logs all alerts for future reference. It strengthens the response time during attempted IP thefts. This module enhances proactive protection and oversight.

5.3 Database Structure

User Registration Table

This table serves as the foundational component of the user authentication and registration system. It contains several fields essential for identifying and managing users: Name, Mobile, Email, Address, UserName, and Password. Each field is of type varchar and is set to NOT NULL to ensure completeness of user data during registration. The id field is of type bigint, set as the PRIMARY KEY, and utilizes auto_increment to uniquely identify each user entry.

To support additional login management, the table includes a Status field that might reflect whether the account is active or inactive, and a LoginKey field that may be used for session validation or tracking login events. The table uses the latin1_swedish_ci collation, which is standard for many MySQL setups. This table is critical for handling secure user access, verifying credentials, and managing user states in a web or application-based environment.

	Field	Type	Collation	Attributes	Null	Default	Extra	Action							
☐	id	bigint(20)			No		auto_increment								
☐	Name	varchar(250)	latin1_swedish_ci		No										
☐	Mobile	varchar(250)	latin1_swedish_ci		No										
☐	Email	varchar(250)	latin1_swedish_ci		No										
☐	Address	varchar(500)	latin1_swedish_ci		No										
☐	UserName	varchar(250)	latin1_swedish_ci		No										
☐	Password	varchar(250)	latin1_swedish_ci		No										
☐	Status	varchar(250)	latin1_swedish_ci		No										
☐	LoginKey	varchar(250)	latin1_swedish_ci		No										

Figure 5.3.1: User Registration Table

File Storage and Key Mapping Table

This table is designed to store detailed and secure information about files uploaded or managed by users in the system. It includes a `UserName` field to associate files with specific users. The `FileInfo` and `FileName` fields store descriptive and identifiable information about each file. To support searchability and classification, the `keyword` field is included.

The table also integrates security by storing public and private key representations in the `Pubhex` and `Privhex` fields, which are used for encryption or secure sharing. Additionally, `Hash1` and `Hash2` fields are used to store hash values, ensuring file integrity and enabling data verification mechanisms. The `id` field acts as the primary key with `auto_increment`, ensuring unique identification for each record. This schema supports strong data security practices and provides a structured method for tracking and verifying user-associated files.

	Field	Type	Collation	Attributes	Null	Default	Extra	Action
☐	<u>id</u>	bigint(20)			No		auto_increment	      
☐	UserName	varchar(250)	latin1_swedish_ci		No			      
☐	FileInfo	varchar(500)	latin1_swedish_ci		No			      
☐	FileName	varchar(250)	latin1_swedish_ci		No			      
☐	keyword	varchar(500)	latin1_swedish_ci		No			      
☐	Pubhex	varchar(500)	latin1_swedish_ci		No			      
☐	Privhex	varchar(500)	latin1_swedish_ci		No			      
☐	Hash1	varchar(500)	latin1_swedish_ci		No			      
☐	Hash2	varchar(500)	latin1_swedish_ci		No			      

Figure 5.3.2: File Storage and Key Mapping Table

CHAPTER 6

SYSTEM DESIGN

6.1 System Architecture

The architecture of the proposed system consists of several key components working together to ensure document security. At the core, Natural Language Processing (NLP) analyzes document content to identify sensitive information, using techniques like TF-IDF for keyword extraction and content classification. Documents are then secured using ECC encryption, ensuring that only authorized users with the correct decryption keys can access them. Additionally, a deception layer is integrated, where fake documents are generated and stored in alternative repositories to mislead unauthorized users. Real-time monitoring tools continuously track interactions with documents, triggering alerts for any unauthorized access or tampering. This layered architecture provides a comprehensive and proactive defense mechanism for safeguarding sensitive information.

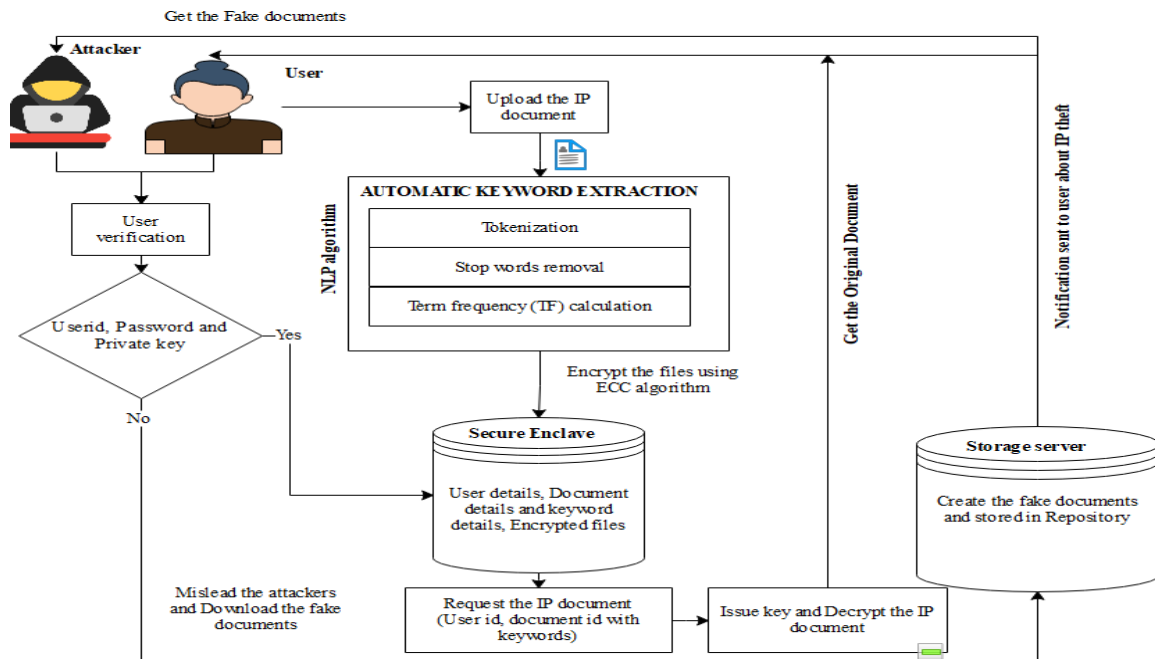


Figure 6.1.1: System architecture

6.2 Workflow Diagram

The workflow diagram illustrates the step-by-step process of securing documents using NLP and encryption. It begins with document upload, followed by content analysis using TF-IDF to detect sensitive information. Sensitive documents are encrypted with ECC encryption and stored securely, while decoy documents are generated to mislead attackers. Access control and monitoring ensure only authorized users interact with real files. Alerts are triggered in real time for any unauthorized activity.

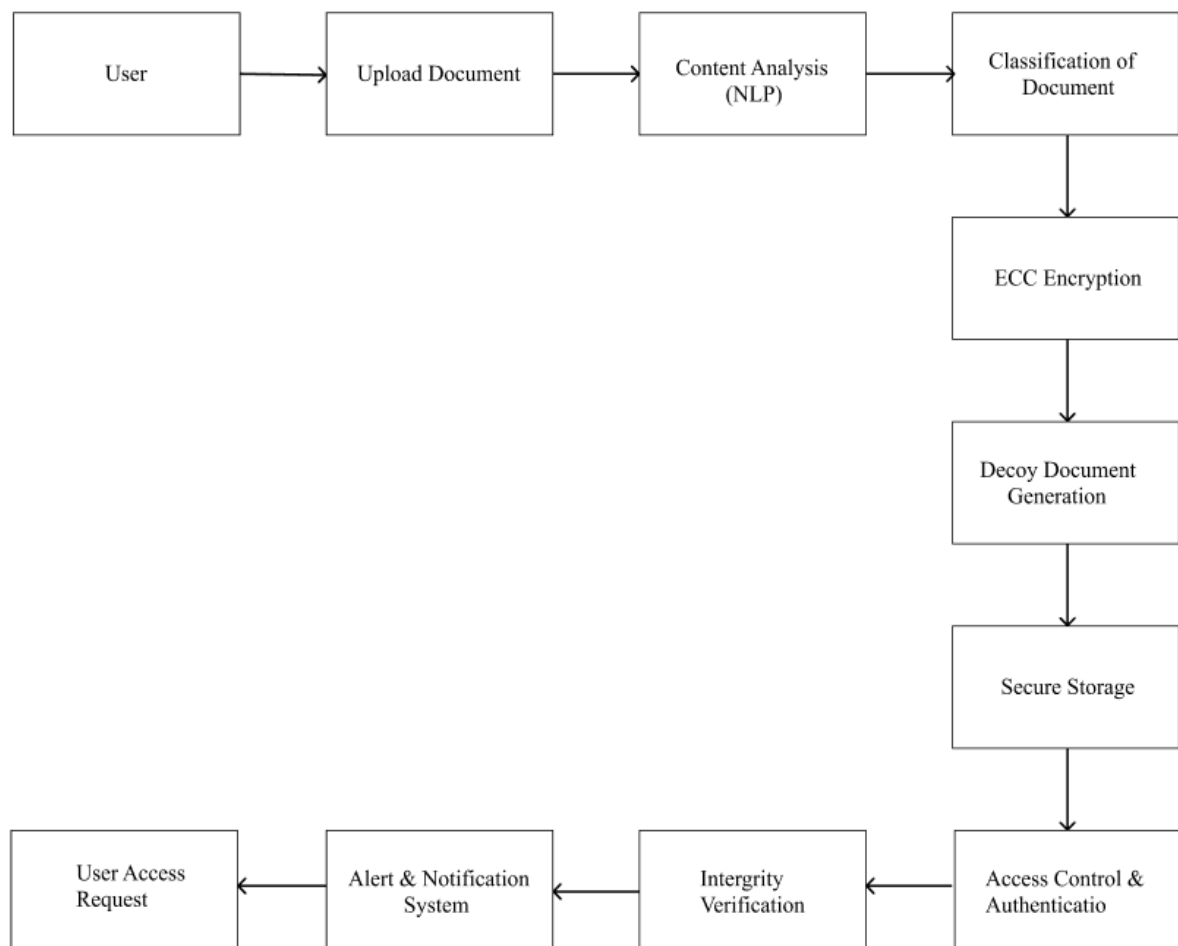


Figure 6.2.1: Workflow diagram

6.3 Use Case Diagram

In its most basic form, a use case diagram is a depiction of a user's interaction with the system that illustrates the connection between the user and the many use cases that the user is involved in. A "system" in this sense refers to something that is being created or run, like a website. The "actors" are individuals or groups functioning inside the system in designated roles.

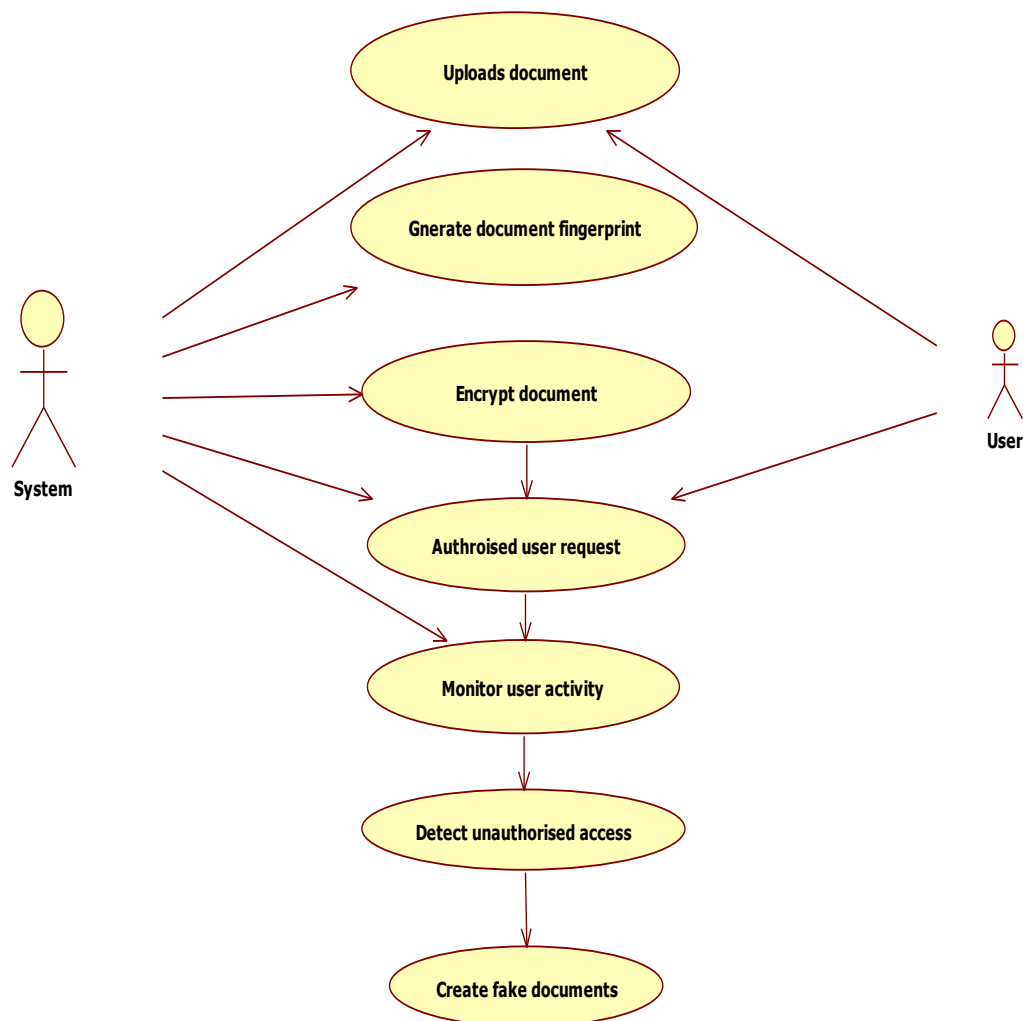


Figure 6.3.1: Use case diagram

6.4 Class Diagram

A class diagram, as defined by the Unified Modeling Language (UML), is a kind of static structural diagram that illustrates a system's classes, properties, functions, and interactions between objects. The fundamental component of object-oriented modeling is the class diagram. It is utilized for both technical modeling which converts the models into computer code and general conceptual modeling of the applications systematic.

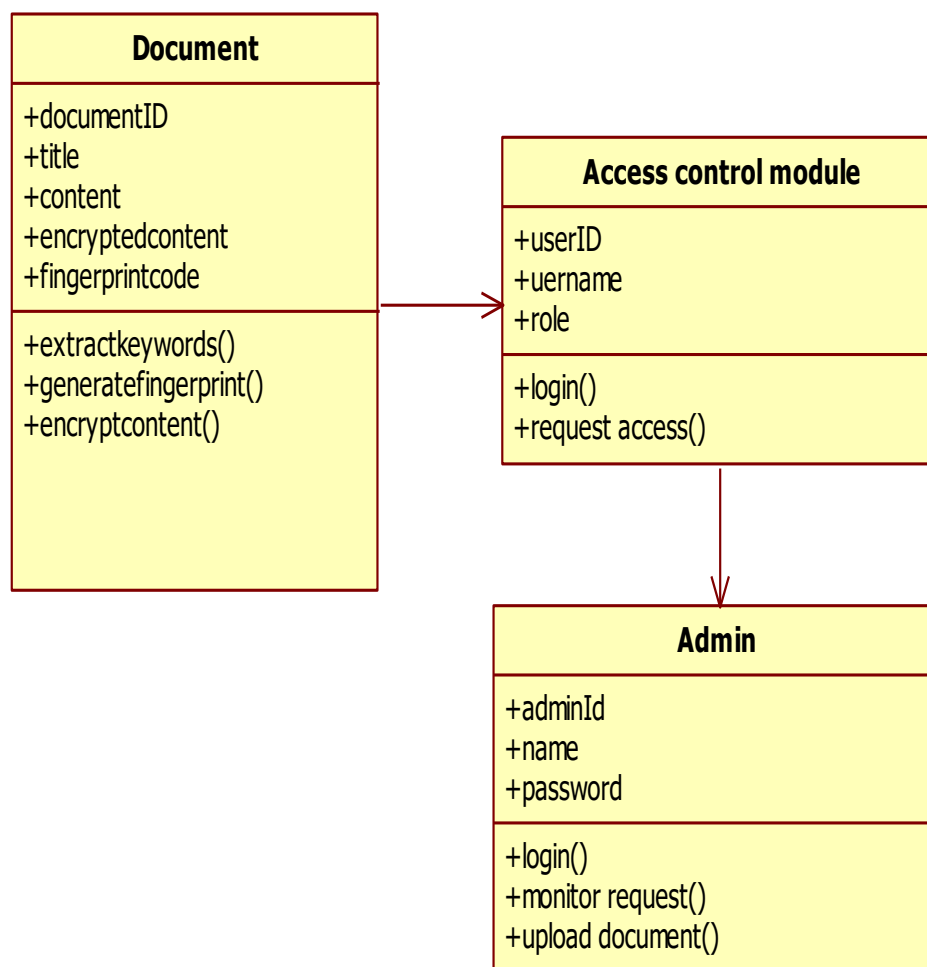


Figure 6.4.1: Class diagram

6.5 Activity Diagram

The activity diagram shows a unique kind of state diagram in which the majority of states are action states and the majority of transitions are brought about by the fulfillment of actions in the source states. One may refer to the action as a system operation. As a result, the control flow is transferred across operations. This flow may occur concurrently, forked, or sequentially. Activity diagrams use a variety of features to address various forms of flow control.

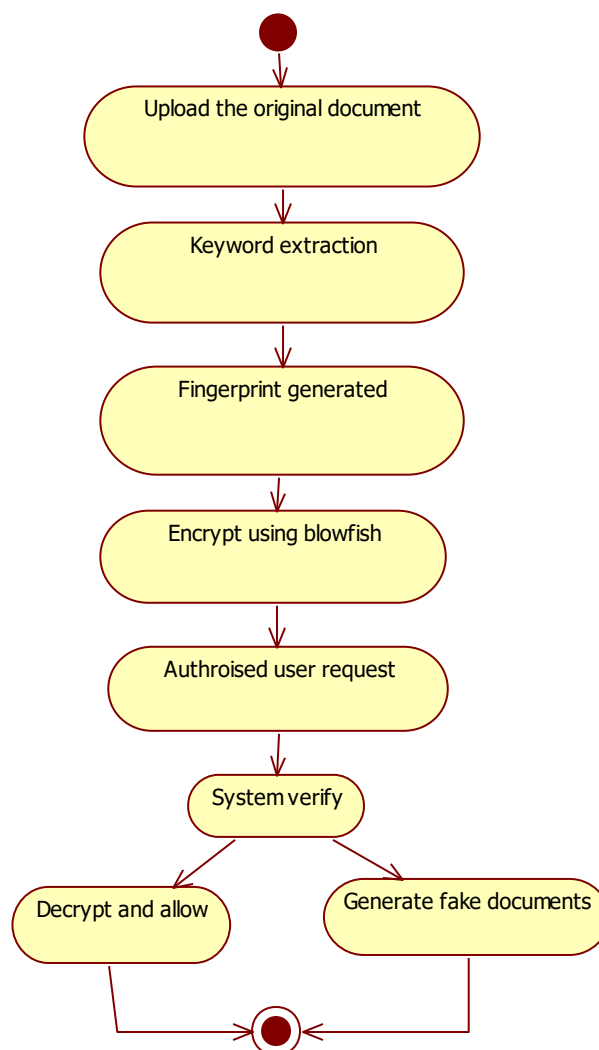


Figure 6.5.1: Activity diagram

6.6 Sequence Diagram

An object's interactions are arranged chronologically in a sequence diagram. It shows the classes and objects that are a part of the scenario as well as the messages that are passed between the objects in order for the scenario to work. Sequence diagrams are commonly linked to the realizations of use cases in the Logical View of the system that is being developed. Event diagrams or event scenarios are other names for sequence diagrams.

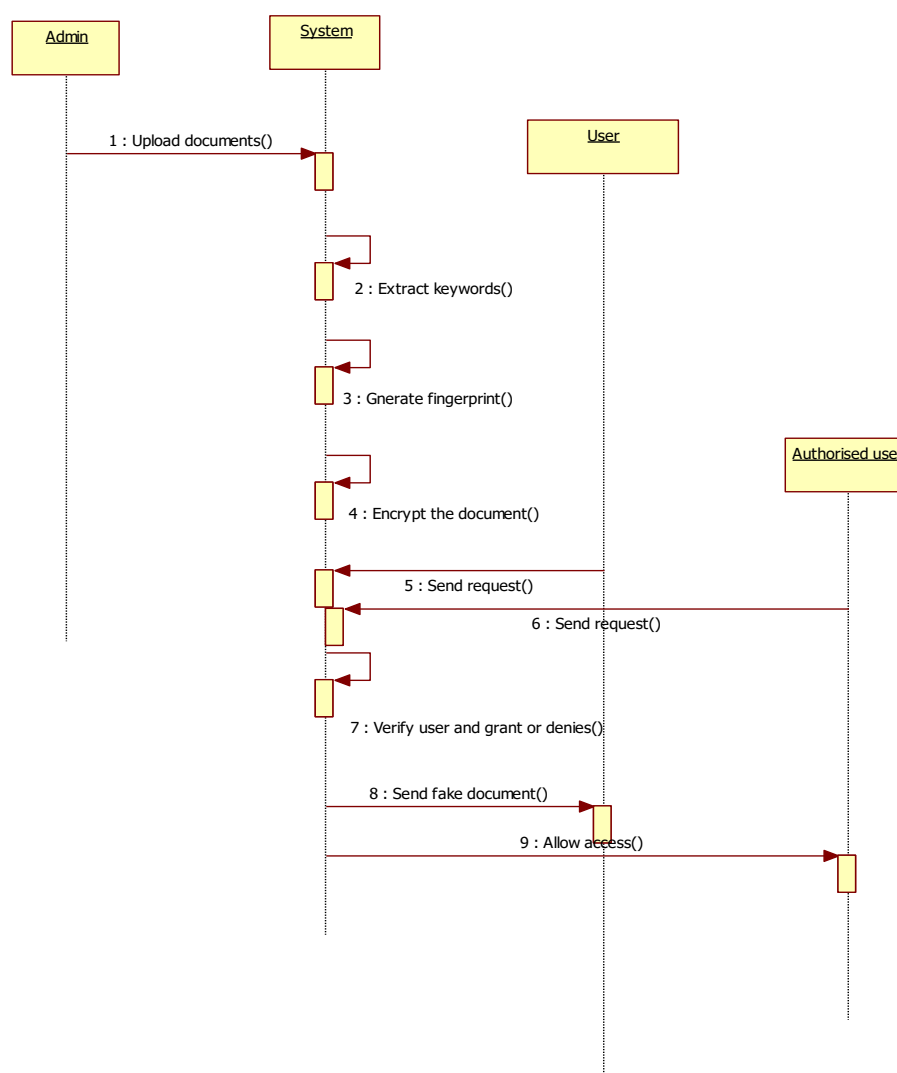


Figure 6.6.1: Sequence diagram

6.7 ER Diagram

An entity relationship diagram (ERD), also known as an entity relationship model, is a graphical representation of an information system that depicts the relationships among people, objects, places, concepts or events within that system. An ERD is a data modeling technique that can help define business processes and be used as the foundation for a relational database. Entity relationship diagrams provide a visual starting point for database design that can also be used to help determine information system requirements throughout an organization. After a relational database is rolled out, an ERD can still serve as a referral point, should any debugging or business process re-engineering be needed later.

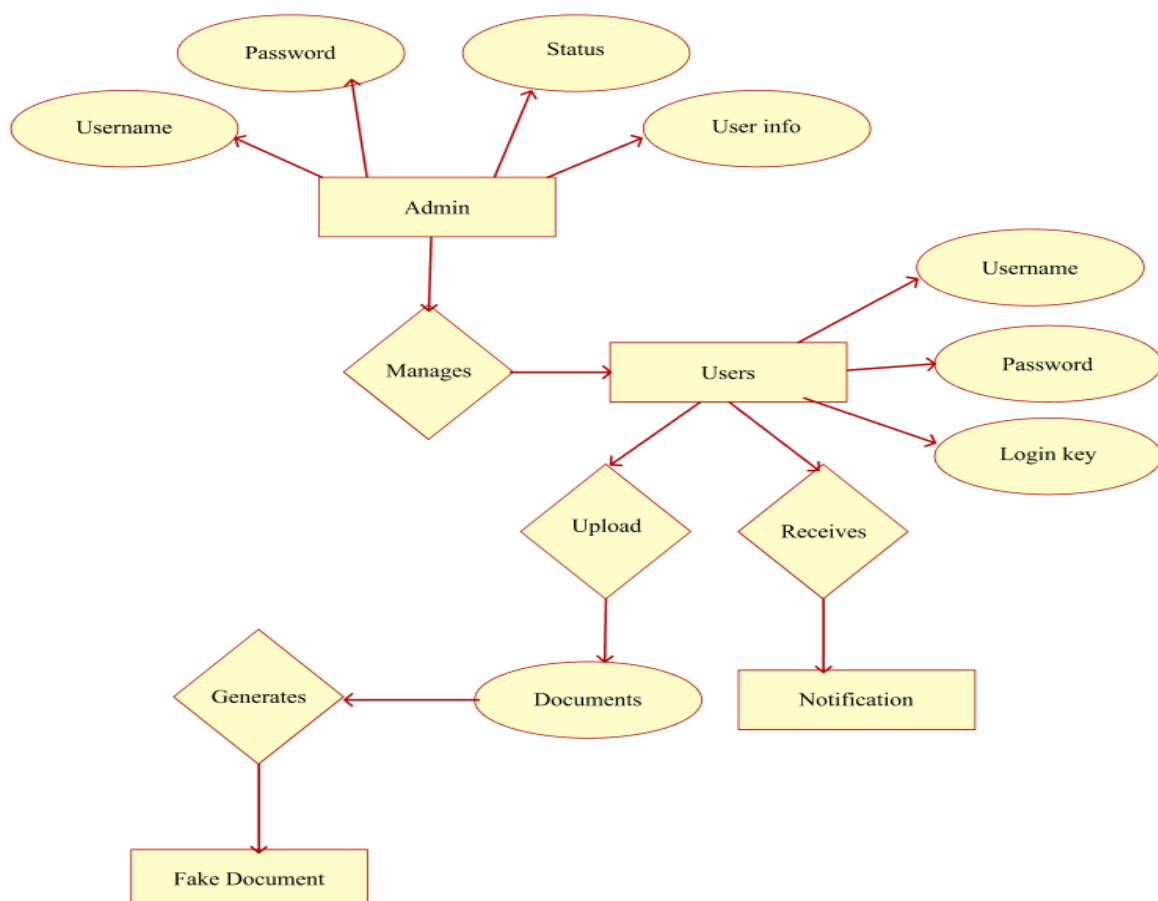


Figure 6.7.1: ER diagram

6.8 Collaboration Diagram

A collaboration diagram often referred to as an interaction diagram or communication diagram uses the Unified Modeling Language (UML) to show the links and interactions between software elements.

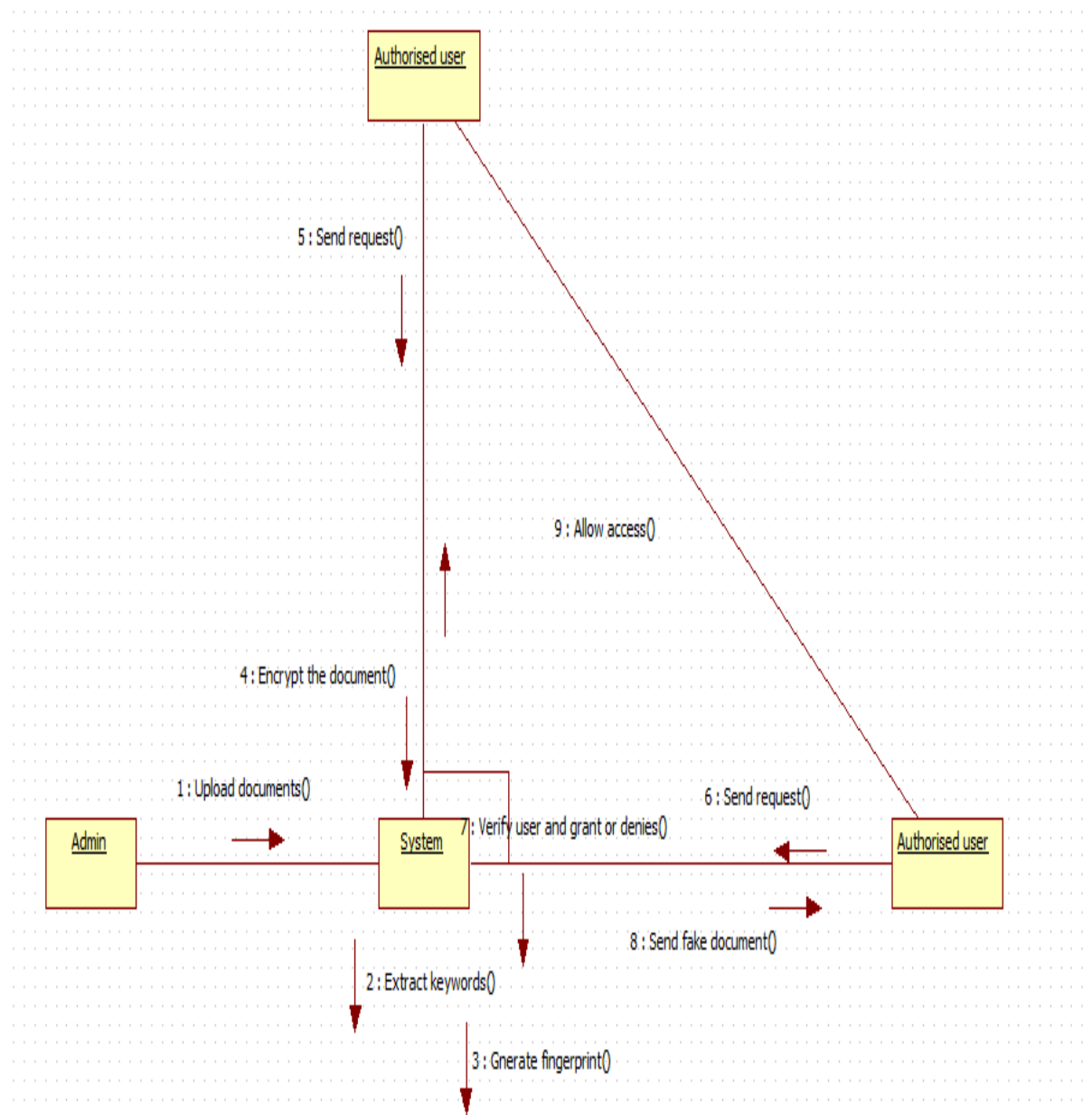


Figure 6.8.1: Collaboration diagram

6.9 Data Flow Diagram

A data flow diagram shows how data is processed within a system based on inputs and outputs. Visual symbols are used to represent the flow of information, data sources and destinations, and where data is stored. Data flow diagrams are often used as a first step toward redesigning a system. They provide a graphical representation of a system at any level of detail, creating an easy-to-understand picture of what the system does. A general overview of a system is represented with a context diagram, also known as a level 0 DFD, which shows a system as a single process. A level 1 diagram provides greater detail, focusing on a system's main functions. Diagrams that are level 2 or higher illustrate a system's functioning with increasing detail. It's rare for a DFD to go beyond level 2 because of the increasing complexity, which makes it less effective as a communication tool.

Level 0

The Level 0 DFD shows how the system is divided into 'sub-systems' (processes), each of which deals with one or more of the data flows to or from an external agent, and which together provide all of the functionality of the system as a whole. It also identifies internal data stores that must be present in order for the system to do its job, and shows the flow of data between the various parts of the system.

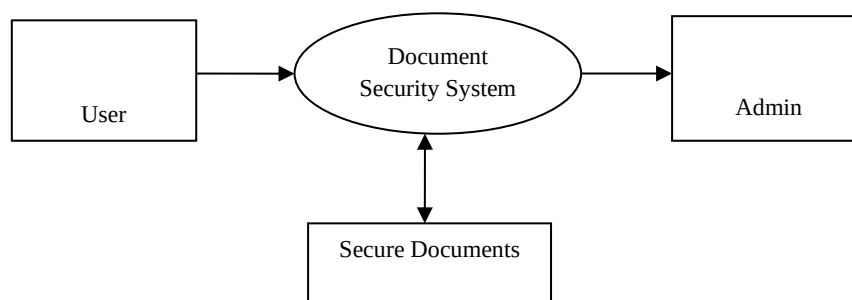


Figure 6.9.1: Level 0- Dataflow diagram

Level 1

The next stage is to create the Level 1 Data Flow Diagram. This highlights the main functions carried out by the system. As a rule, to describe the system was using between two and seven functions - two being a simple system and seven being a complicated system. This enables us to keep the model manageable on screen or paper.

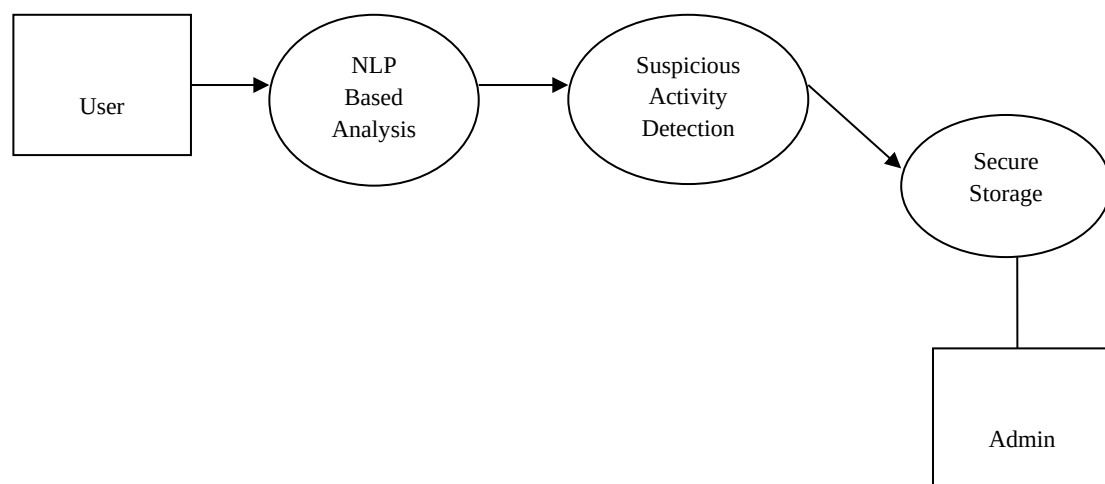


Figure 6.9.2: Level 1- Dataflow diagram

Level 2

A Data Flow Diagram (DFD) tracks processes and their data paths within the business or system boundary under investigation. A DFD defines each domain boundary and illustrates the logical movement and transformation of data within the defined boundary. The diagram shows 'what' input data enters the domain, 'what' logical processes the domain applies to that data, and 'what' output data leaves the domain. Essentially, a DFD is a tool for process modeling and one of the oldest.

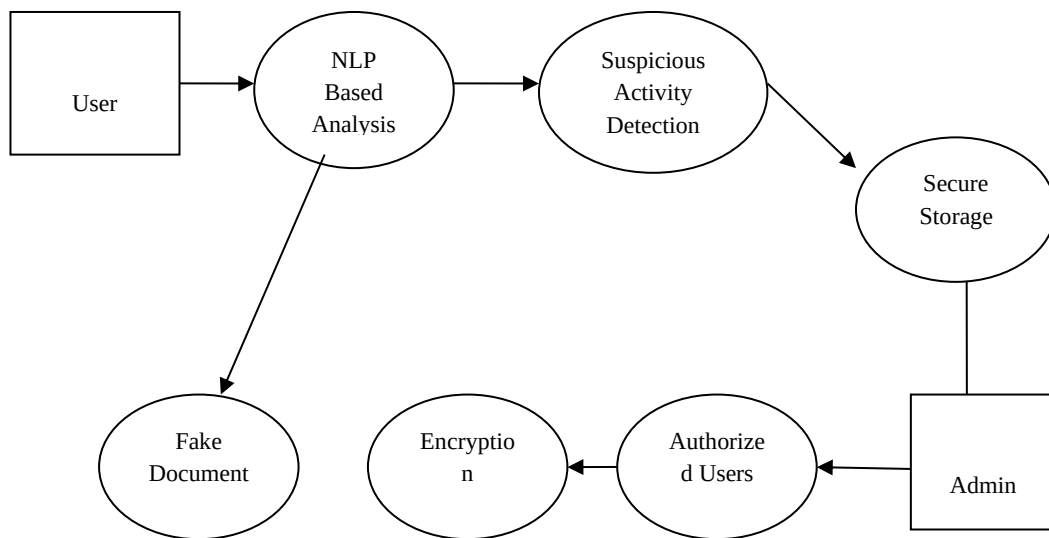


Figure 6.9.3: Level 2- Dataflow diagram

CHAPTER 7

SYSTEM TESTING

Testing is a set activity that can be planned and conducted systematically. Testing begins at the module level and work towards the integration of entire computers based system. Nothing is complete without testing, as it is vital success of the system.

Testing Objectives:

There are several rules that can serve as testing objectives, they are

1. Testing is a process of executing a program with the intent of finding an error.
2. A good test case is one that has high probability of finding an undiscovered error.
3. A successful test is one that uncovers an undiscovered error.

If testing is conducted successfully according to the objectives as stated above, it would uncover errors in the software. Also testing demonstrates that software functions appear to be working according to the specification, that performance requirements appear to have been met.

There are three ways to test a program

1. For Correctness
2. For Implementation efficiency
3. For Computational Complexity.

Tests used for implementation efficiency attempt to find ways to make a correct program faster or use less storage. It is a code-refining process, which reexamines

the implementation phase of algorithm development. Tests for computational complexity amount to an experimental analysis of the complexity of an algorithm or an experimental comparison of two or more algorithms, which solve the same problem.

The data is entered in all forms separately and whenever an error occurred, it is corrected immediately. A quality team deputed by the management verified all the necessary documents and tested the Software while entering the data at all levels.

Types Of Testing

The development process involves various types of testing. Each test type addresses a specific testing requirement. The most common types of testing involved in the development process are:

- Unit Test
- Functional Test
- Integration Test
- White box Test
- Black box Test
- System Test
- Validation Test
- Acceptance Test

Unit Testing

The first test in the development process is the unit test. The source code is normally divided into modules, which in turn are divided into smaller units

called units. These units have specific behavior. The test done on these units of code is called unit test. Unit test depends upon the language on which the project is developed. Unit tests ensure that each unique path of the project performs accurately to the documented specifications and contains clearly defined inputs and expected results.

Functional Testing

Functional test can be defined as testing two or more modules together with the intent of finding defects, demonstrating that defects are not present, verifying that the module performs its intended functions as stated in the specification and establishing confidence that a program does what it is supposed to do.

Integration Testing

In integration testing modules are combined and tested as a group. Modules are typically code modules, individual applications, source and destination applications on a network, etc. Integration Testing follows unit testing and precedes system testing. Testing after the product is code complete. Betas are often widely distributed or even distributed to the public at large in hopes that they will buy the final product when it is released.

White Box Testing

Testing based on an analysis of internal workings and structure of a piece of software. This testing can be done using the percentage value of load and energy. The tester should know what exactly is done in the internal program. It includes techniques such as Branch Testing and Path Testing. White box testing also called as Structural Testing or Glass Box Testing.

Black Box Testing

In block box testing without knowledge of the internal workings of the item being tested. Tests are usually functional. This testing can be done by the user who has no knowledge of how the shortest path is found.

System Testing

System testing is defined as testing of a complete and fully integrated software product. This testing falls in black-box testing wherein knowledge of the inner design of the code is not a pre-requisite and is done by the testing team. It is the final test to verify that the product to be delivered meets the specifications mentioned in the requirement document. It should investigate both functional and non-functional requirements.

Validation Testing

The process of evaluating software during the development process or at the end of the development process to determine whether it satisfies specified business requirements. Validation Testing ensures that the product actually meets the client's needs. It can also be defined as to demonstrate that the product fulfills its intended use when deployed on appropriate environment.

Acceptance Testing

This is a type of testing done by users, customers, or other authorised entities to determine application/software needs and business processes. Acceptance testing is the most important phase of testing as this decides whether the client approves the application/software or not. It may involve functionality, usability, performance, and U.I of the application. It is also known as user acceptance testing (UAT), operational acceptance testing (OAT), and end-user testing.

Software Testing Strategies

In order to make sure that the programme satisfies its intended criteria and performs as expected, software testing strategies are detailed blueprints that direct the testing procedure throughout a software development project. Various testing approaches are used, depending on the particulars of the project:

Waterfall Testing Strategy: The requirements, design, implementation, testing, and deployment phases of the waterfall model are sequentially approached, with each stage being finished before going on to the next. With a focus on validation against predetermined requirements, this strategy's testing is thorough and well-documented.

V-Model Testing Strategy: The V-Model, which is a development and testing phase relationship model, is an extension of the Waterfall model. To guarantee that requirements are verified early in the process, there is a testing phase that corresponds with each development phase.

Iterative Testing Strategy: This approach, which incorporates incremental testing with gradual additions of modules or components, is perfect for iterative development. Early flaw detection and adaptable modifications in response to feedback are made possible by it.

Exploratory Testing Strategy: Without preset test cases, testers use their experience and instincts to investigate the product, which is a helpful method for identifying bugs that scripted tests, could miss.

Automated Testing Strategy: This approach is especially helpful for regression testing, repetitive jobs, and large-scale projects since it emphasises the use of automated testing tools and scripts to carry out test cases.

The nature of the project, the development process, the available funds, and the amount of time all play a role in selecting a testing approach.

CHAPTER 8

CONCLUSION AND FUTURE ENHANCEMENTS

Conclusion

In conclusion, this project offers a robust and intelligent system for protecting Intellectual Property (IP) documents from unauthorized access, tampering, and theft. By integrating advanced technologies such as Natural Language Processing (NLP), ECC encryption, and secure user authentication methods, the system ensures that only verified users can access or modify sensitive content. The automatic keyword extraction and document fingerprinting enhance content monitoring, allowing the system to detect suspicious changes or data leakage effectively. Furthermore, the deployment of decoy documents misleads potential attackers and prevents them from obtaining the original sensitive data, adding another layer of proactive security. Additionally, the system's modular structure enables seamless integration of user management, document tracking, and real-time alert mechanisms.

The implementation of multi-factor authentication and detailed logging of user activity enhances accountability and allows for accurate auditing. The notification module keeps users informed of any suspicious activities, reinforcing user awareness and security. Overall, this project represents a comprehensive solution to modern digital threats faced by IP document holders, combining automation, cryptographic protection, and intelligent analysis to ensure privacy, integrity, and secure document sharing in an increasingly connected world. It not only deters malicious insiders and external attackers but also fosters a culture of accountability and vigilance. The system's adaptability makes it suitable for various domains where IP protection is crucial, including legal, corporate, and academic environments.

Future Enhancements

- **Integration of AI-Based Threat Detection**

Implement advanced AI and machine learning models to predict and prevent potential security breaches by analyzing user behavior patterns and access logs.

- **Blockchain-Based Document Logging**

Use blockchain technology to create immutable logs of document access and modifications, enhancing transparency, traceability, and tamper-proof auditing.

- **Cloud Storage and Remote Access**

Enable secure cloud-based document storage with end-to-end encryption to support remote access and collaboration without compromising security.

- **Support for Multiple File Formats**

Extend support for various document types including spreadsheets, presentations, multimedia files, and code repositories, providing broader coverage.

- **Mobile Application Integration**

Develop a companion mobile app for real-time notifications, quick document access, and biometric authentication on-the-go, improving user convenience and accessibility.

CHAPTER 9

APPENDIX – 1

SOURCE CODE

```
from flask import Flask, render_template, flash, request, session, send_file,
jsonify

from flask import render_template, redirect, url_for, request

import mysql.connector

from ecies.utils import generate_key

from ecies import encrypt, decrypt

import base64, os, sys

import nltk

from nltk.corpus import stopwords

from nltk.tokenize import word_tokenize

from collections import Counter

import string

import docx # for reading .docx files

import fitz # PyMuPDF, for reading .pdf files


# Ensure necessary NLTK packages are downloaded

nltk.download('punkt')

nltk.download('stopwords')


app = Flask(__name__)

app.config['DEBUG']
```

```
app.config['SECRET_KEY'] = '7d441f27d441f27567d441f2b6176a'
```

```
@app.route("/")
```

```
def homepage():
```

```
    return render_template('index.html')
```

```
@app.route("/ServerLogin")
```

```
def ServerLogin():
```

```
    return render_template('ServerLogin.html')
```

```
@app.route('/NewUser')
```

```
def NewUser():
```

```
    return render_template('NewUser.html')
```

```
@app.route('/UserLogin')
```

```
def UserLogin():
```

```
    return render_template('UserLogin.html')
```

```
@app.route("/serverlogin", methods=['GET', 'POST'])
```

```
def serverlogin():
```

```
    if request.method == 'POST':
```

```
        if request.form['uname'] == 'server' and request.form['password'] ==  
'server':
```

```
            cur = conn.cursor()
```

```
            cur.execute("SELECT * FROM regtb where status='waiting'")
```

```

        data = cur.fetchall()

        conn = mysql.connector.connect(user='root', password="",
host='localhost', database='1fakedocclouddb')

        cur = conn.cursor()

        cur.execute("SELECT * FROM regtb where status='Active'")

        data1 = cur.fetchall()

        return render_template('ServerHome.html', data=data, data1=data1)

    else:

        flash('Username or Password is wrong')

        return render_template('ServerLogin.html')

```

```

@app.route("/ServerHome")

```

```

def ServerHome()

```

```

    conn = mysql.connector.connect(user='root', password="", host='localhost',
database='1fakedocclouddb')

    cur = conn.cursor()

    cur.execute("SELECT * FROM regtb where status !='waiting'")

    data1 = cur.fetchall()

    return render_template('ServerHome.html', data=data, data1=data1)

```

```

@app.route("/Approved")

```

```

def Approved():

```

```

    id = request.args.get('lid')

    email = request.args.get('email')

    import random

```

```

loginkey = random.randint(1111, 9999)

message = "Your Defense Login Key :" + str(loginkey)

sendmail(email, message)

conn = mysql.connector.connect(user='root', password="", host='localhost',
database='1fakedocclouddb')

cursor = conn.cursor()

```

```
@app.route("/Reject")
```

```
def Reject():
```

```

    id = request.args.get('lid')

    email = request.args.get('email')

    message = "Your Request Rejected"

    sendmail(email, message)

    conn = mysql.connector.connect(user='root', password="", host='localhost',
database='1fakedocclouddb')

    cur = conn.cursor()

    cur.execute("SELECT * FROM regtb where status='waiting'")

    return render_template('ServerHome.html', data=data, data1=data1)

```

```
@app.route("/SFileInfo")
```

```
def SFileInfo():
```

```

    conn = mysql.connector.connect(user='root', password="", host='localhost',
database='1fakedocclouddb')

    cur = conn.cursor()

    cur.execute("SELECT * FROM filetb ")

```



```

data = cur.fetchall()

return render_template('SFileInfo.html', data=data)

@app.route("/newuser", methods=['GET', 'POST'])
def newuser():
    if request.method == 'POST':
        uname = request.form['uname']
        mobile = request.form['mobile']
        email = request.form['email']
        address = request.form['address']
        username = request.form['username']
        password = request.form['password']
        if data is None:
            conn = mysql.connector.connect(user='root', password="",
            host='localhost', database='1fakedocclouddb')

            cursor = conn.cursor()

            cursor.execute(
                "INSERT INTO regtb VALUES ('" + uname + "','" + mobile + "','" +
email + "','" + address + "','" + username + "','" + password + "','waiting',)")

            conn.commit()

            conn.close()

            flash('Record Saved!')

            return render_template('NewUser.html')
        else:
            flash('Already Register This UserName!')

```

```

        return render_template('NewUser.html')

@app.route("/userlogin", methods=['GET', 'POST'])
def userlogin():
    if request.method == 'POST':
        username = request.form['uname']
        password = request.form['password']
        loginkey = request.form['loginkey']
        session['uname'] = request.form['uname']
        if data is None:
            #flash('Username or Password is wrong')

            conn = mysql.connector.connect(user='root', password="",
host='localhost', database='1fakedocclouddb')

            cur = conn.cursor()
        else:
            Status = data[7]
            lkey = data[8]
            session['email'] = data[3]
            if Status == "waiting":
                flash('Waiting For Server Approved!')
                return render_template('UserLogin.html')
            else:
                if lkey == loginkey:
                    conn = mysql.connector.connect(user='root', password="",
host='localhost',database='1fakedocclouddb')

```

```

        flash('Login Successfully')
        return render_template('UserHome.html', data=data1)
    else:
        flash('Login Key Incorrect')
        return render_template('UserLogin.html')

@app.route("/UserHome")
def UserHome():
    return render_template('ServerHome.html', data=data)

@app.route("/UserFileUpload")
def UserFileUpload():
    return render_template('UserFileUpload.html', uname=session['uname'])

import hmac
import hashlib
import binascii

def create_sha256_signature(key, message):
    byte_key = binascii.unhexlify(key)
    message = message.encode()
    return hmac.new(byte_key, message, hashlib.sha256).hexdigest().upper()

# Function to read text from .docx files

```

```

def read_docx(file_path):
    doc = docx.Document(file_path)
    text = '\n'.join([paragraph.text for paragraph in doc.paragraphs])
    return text

# Function to read text from .pdf files
def read_pdf(file_path):
    text = ""
    with fitz.open(file_path) as pdf:
        for page_num in range(pdf.page_count):
            page = pdf[page_num]
            text += page.get_text()
    return text

# Calculate term frequency (TF)
term_frequency = Counter(words)
top_terms = term_frequency.most_common(5)
return top_terms

def read_text_file(file_path):
    try:
        with open(file_path, 'r', encoding='utf-8') as file: # specify UTF-8
            encoding
            text = file.read()
    except UnicodeDecodeError:

```

```

        # Fallback if UTF-8 fails, using a more flexible encoding
        with open(file_path, 'r', encoding='ISO-8859-1') as file:
            text = file.read()

    return text

from docx import Document
from fpdf import FPDF
import random

def generate_random_paragraphs(num_paragraphs=3):
    # Replace with your desired random paragraph generator
    sample_paragraphs = [
        "Lorem ipsum dolor sit amet, consectetur adipiscing elit. Quisque interdum, urna vel fringilla hendrerit, est metus egestas nunc, in commodo elit lorem sed arcu.",
        "Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Pellentesque sit amet consectetur quam. Integer scelerisque risus id nisi aliquet, a suscipit mi congue.",
        "Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Nulla facilisi. Aenean posuere lacus id volutpat accumsan.",
        "Maecenas eget risus sed felis consequat gravida. Etiam vel lorem vitae risus posuere dictum. In hac habitasse platea dictumst.",
    ]

    return "\n\n".join(random.choices(sample_paragraphs, k=num_paragraphs))

def write_to_txt(content, file_path):
    with open(file_path, 'w') as file:

```

```

        file.write(content)

def write_to_docx(content, file_path):
    doc = Document()
    for paragraph in content.split('\n\n'):
        doc.add_paragraph(paragraph)
    doc.save(file_path)

# Main function to handle both file types
def process_file(file_path, fname):
    # Determine file type and read accordingly
    if file_path.endswith('.docx'):
        write_to_docx(new_content, output_pdf)
    elif file_path.endswith('.txt'):
        write_to_txt(new_content, output_pdf)
    else:
        raise ValueError("Unsupported file format. Only .docx ,.txt and .pdf are supported.")

@app.route("/upload", methods=['GET', 'POST'])
def upload():
    if request.method == 'POST':
        username = session['username']
        info = request.form['info']
        file = request.files['file']

```

```

privhex = secp_k.to_hex()
pubhex = secp_k.public_key.format(True).hex()
filepath = "./static/upload/" + savename
head, tail = os.path.split(filepath)
# file_path = '333.pdf' # or 'path_to_your_file.pdf'
tf = process_file(filepath, savename)
keywo = "
mailkey = "
    if keywo == ":
        keywo = word
    else:
        keywo = keywo + " " + word
    if mailkey == "":
        mailkey = word + " Frequency:" + str(frequency)
    else:
        mailkey = mailkey + " " + word + " Frequency:" + str(frequency)
    return render_template('OwnerFileUpload.html')
else:
    print("Binary of the file:", data)
    encrypted_secp = encrypt(pubhex, data)
    print("Encrypted binary:", encrypted_secp)
    if data2:
        da = cursor1.fetchone()
        if da:

```

```

        d = da[0]

        print(d)

        flash('File Upload And Encrypt Successfully ')

        return render_template('UserFileUpload.html', pkey=privhex,
uname=uname, tf=tf)

    else:

        hash1 = '0'

        num1 = random.randrange(1111, 9999)

        conn.close()

        flash('File Upload And Encrypt Successfully ')

        if da:

            d = da[0]

            return render_template('UserFileUpload.html', pkey=privhex,
uname=uname, tf=tf)

@app.route('/UFileInfo')
def UFileInfo():

    cur.execute("SELECT * FROM filetb where username='" + session['uname']
+ "'")

    data1 = cur.fetchall()

    return render_template('UFileInfo.html', data=data1)

@app.route('/USearch')
def USearch():

    return render_template('USearch.html')

```



```

@app.route("/search", methods=['GET', 'POST'])
def search():
    if request.method == 'POST':
        fname = request.form['uname']
        uname = session['uname']
        # flash('Login Successfully')
        return render_template('USearch.html', data=data1)

```

```

@app.route("/search1", methods=['GET', 'POST'])
def search1():
    if request.method == 'POST':
        fname = request.form['uname']
        data1 = cur.fetchall()
        # flash('Login Successfully')
        return render_template('UserHome1.html', data=data1)

```

```

@app.route("/Download1")
def Download1():
    lid = request.args.get('lid')
    data = cursor.fetchone()
    if data:
        fname = data[3]
        uname = data[1]
        return send_file(newfilepath1, as_attachment=True)

```

```

@app.route("/Download")
def Download():
    lid = request.args.get('lid')
    if data:
        return send_file(newfilepath1, as_attachment=True)
def sendmail(Mailid, message):
    # storing the subject
    # start TLS for security
    s.starttls()
    # Authentication
    s.login(fromaddr, "qmgn xecl bkqv musr")
    # Converts the Multipart msg into a string
    text = msg.as_string()
    # sending the mail
    s.sendmail(fromaddr, toaddr, text)
    # terminating the session
    s.quit()

if __name__ == '__main__':
    app.run(debug=True, use_reloader=True)

```

APPENDIX – 2

SCREENSHOT

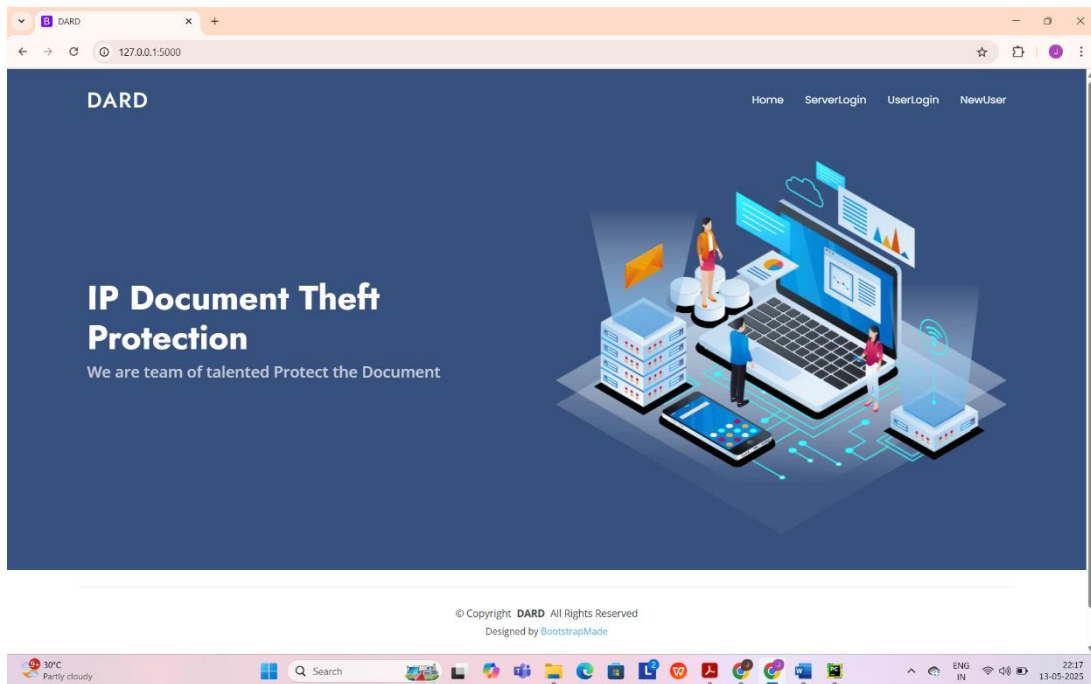


Figure A.9.1: Home page

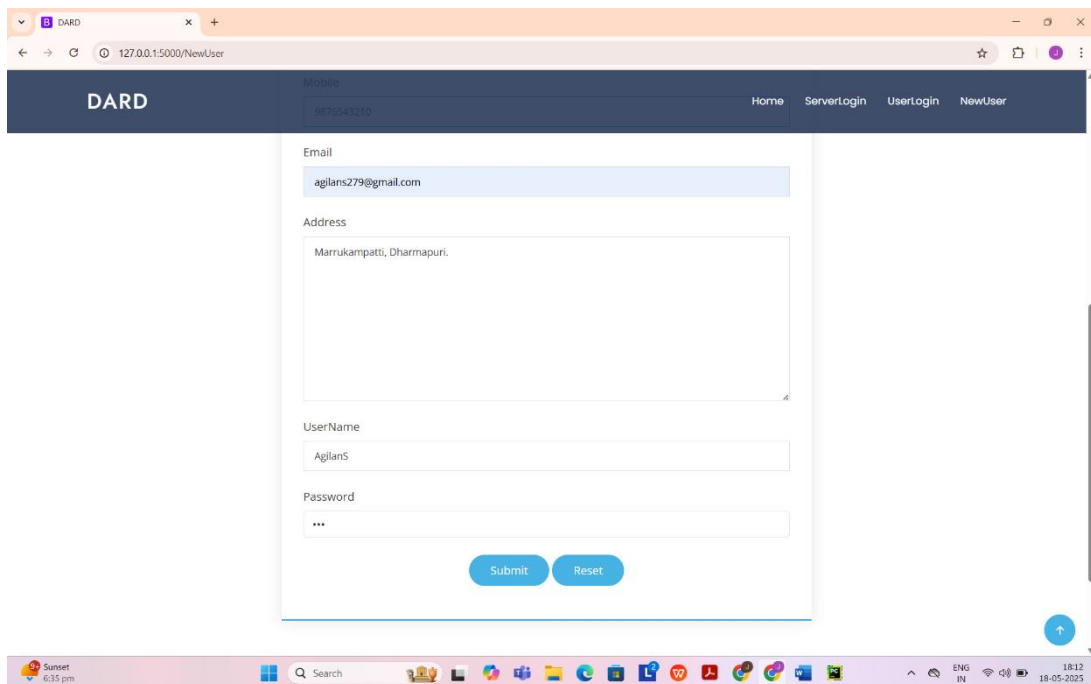


Figure A.9.2: New User Login Page

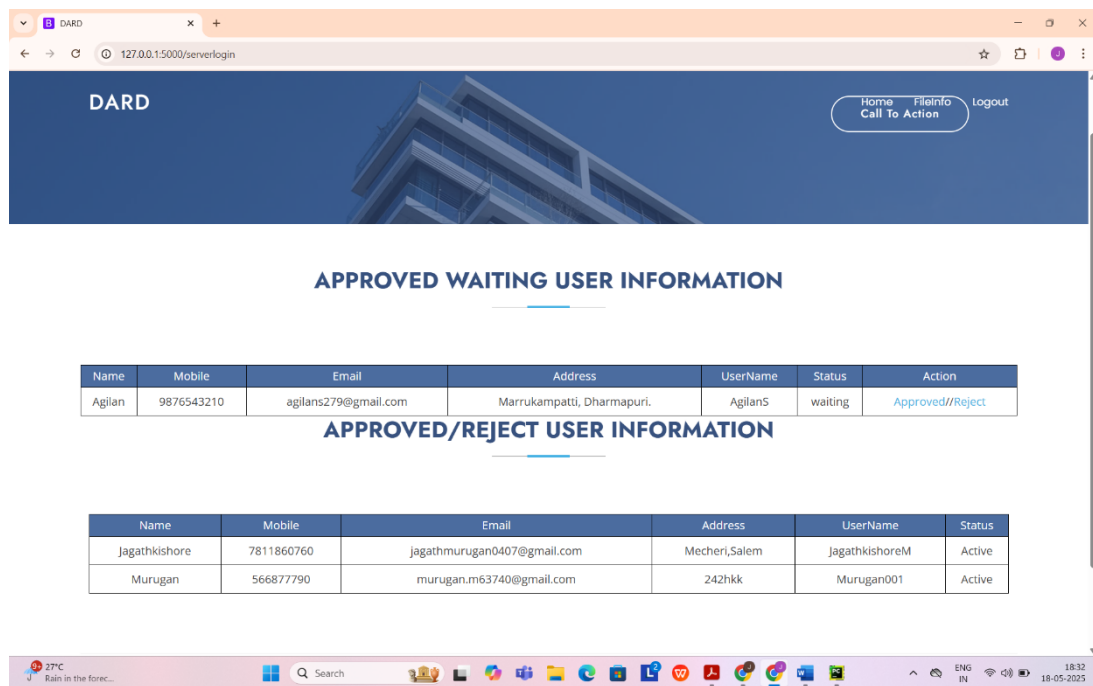


Figure A.9.3: User Approval page

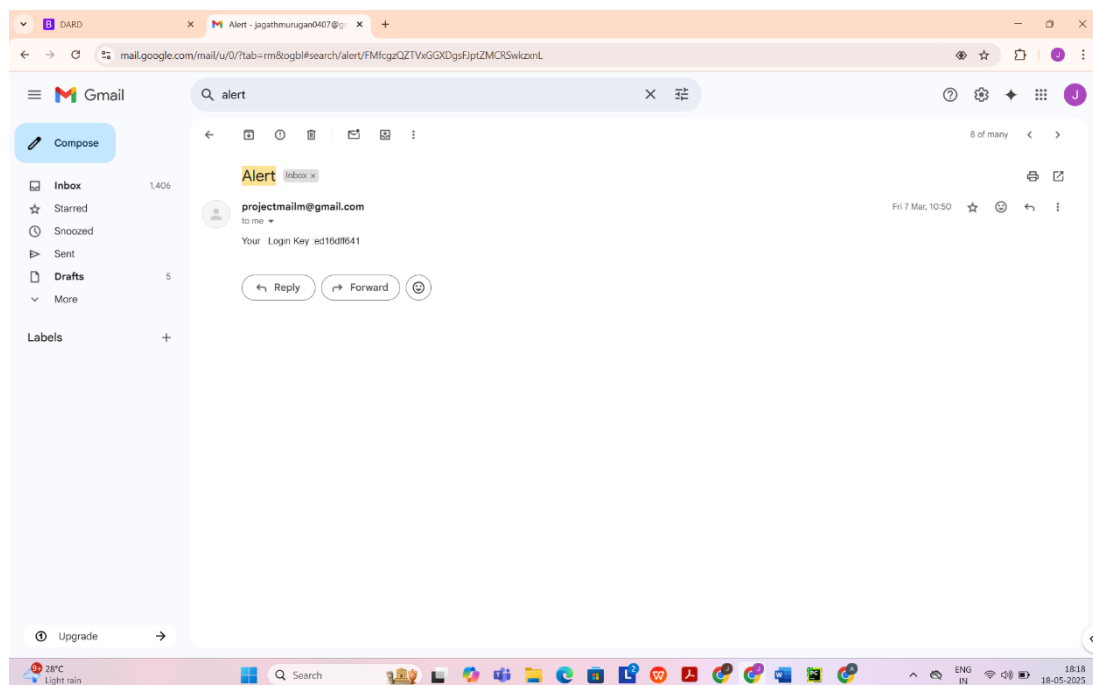


Figure A.9.4: Login Key Page

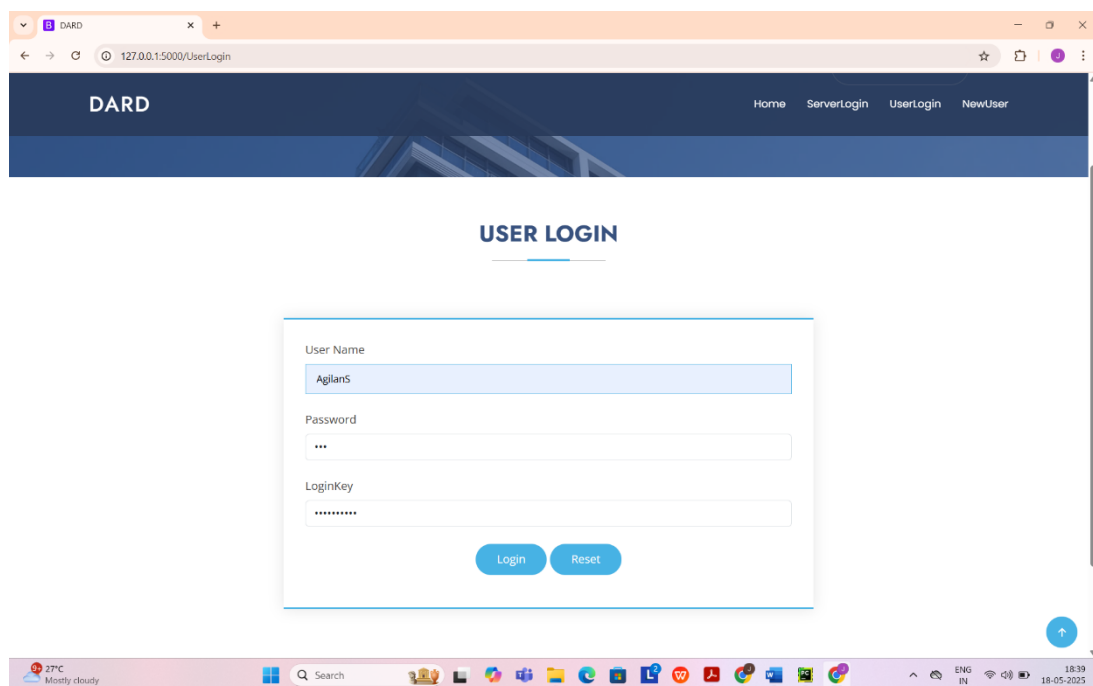


Figure A.9.5: User Login Page

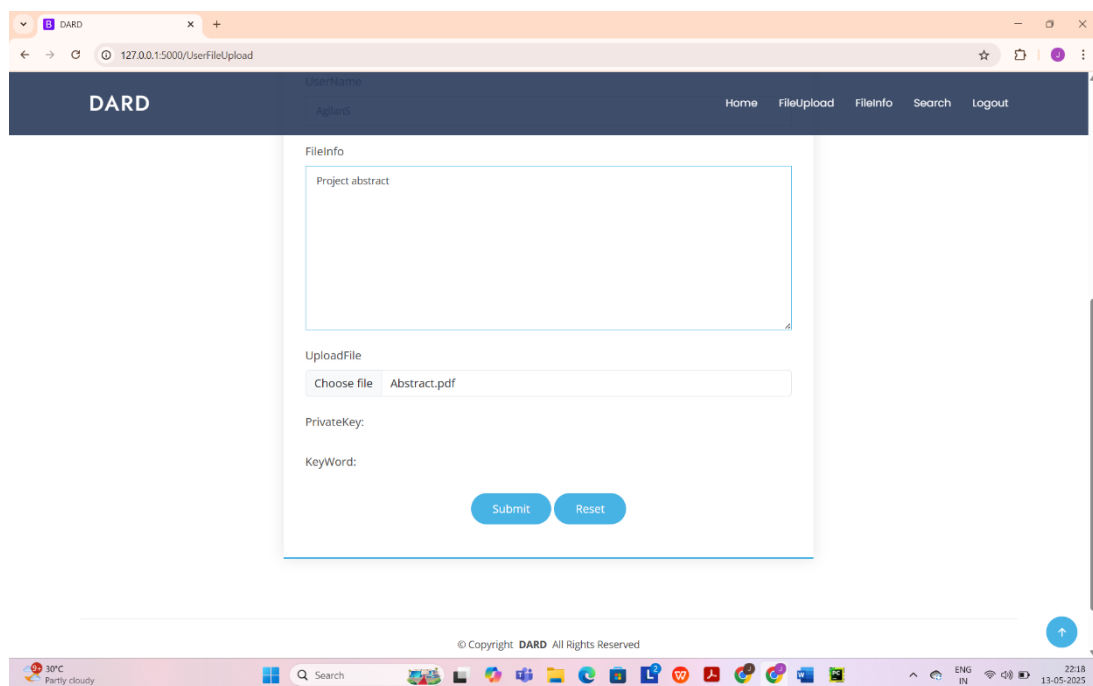


Figure A.9.6: Document Upload Page

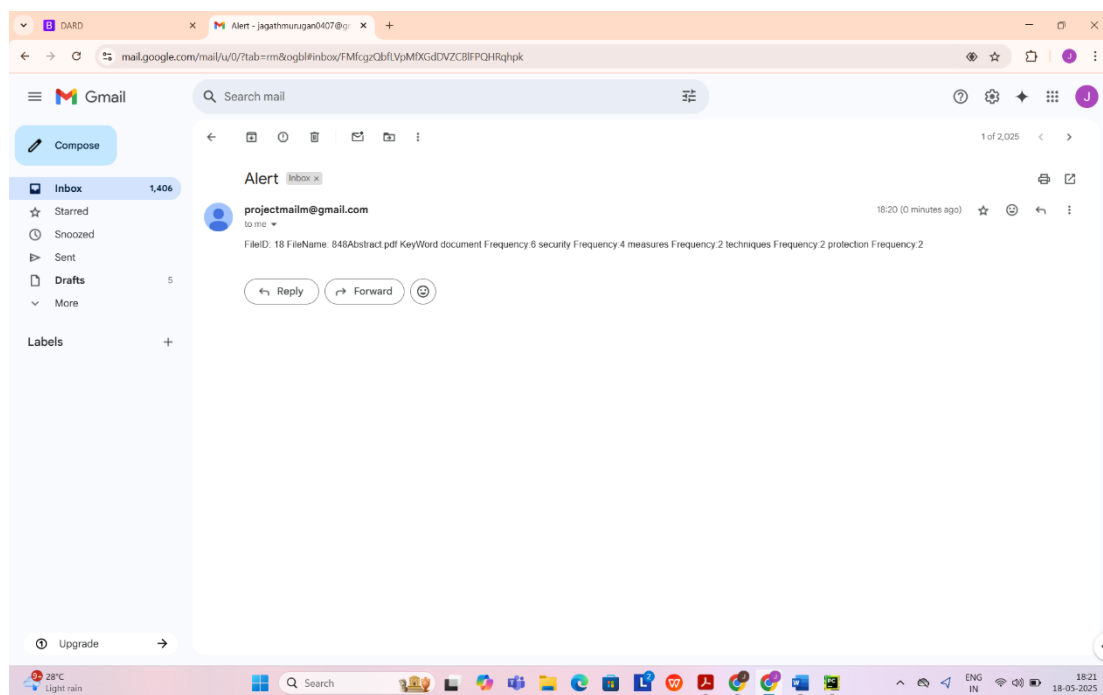


Figure A.9.7: Document Keyword Page

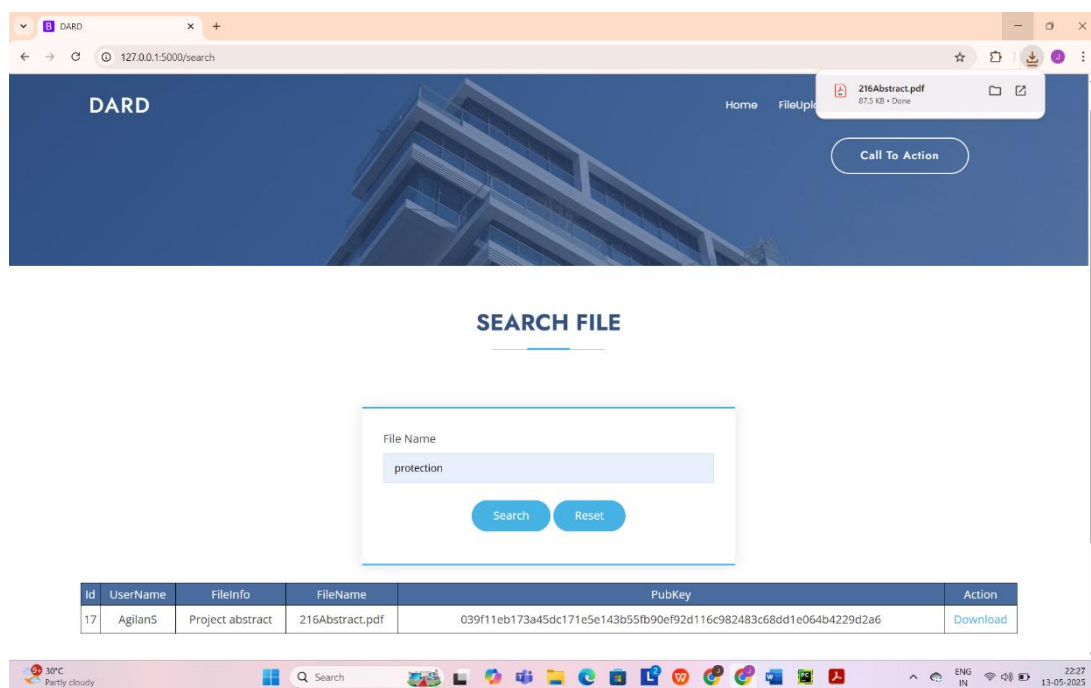


Figure A.9.8: Document Download page

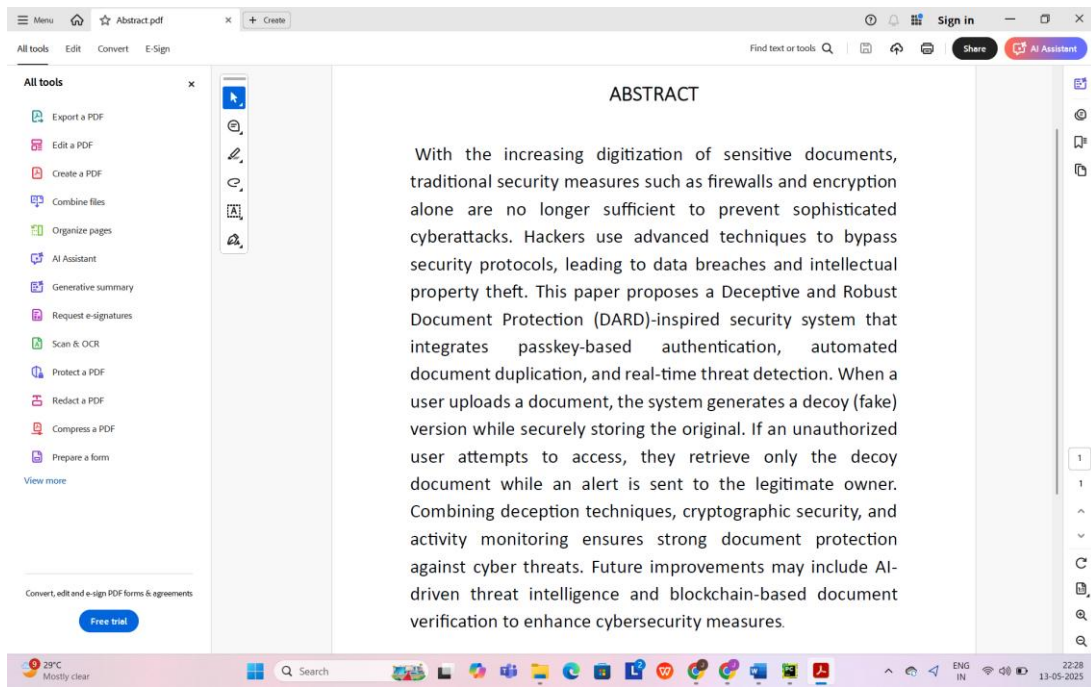


Figure A.9.9: Original Document Page

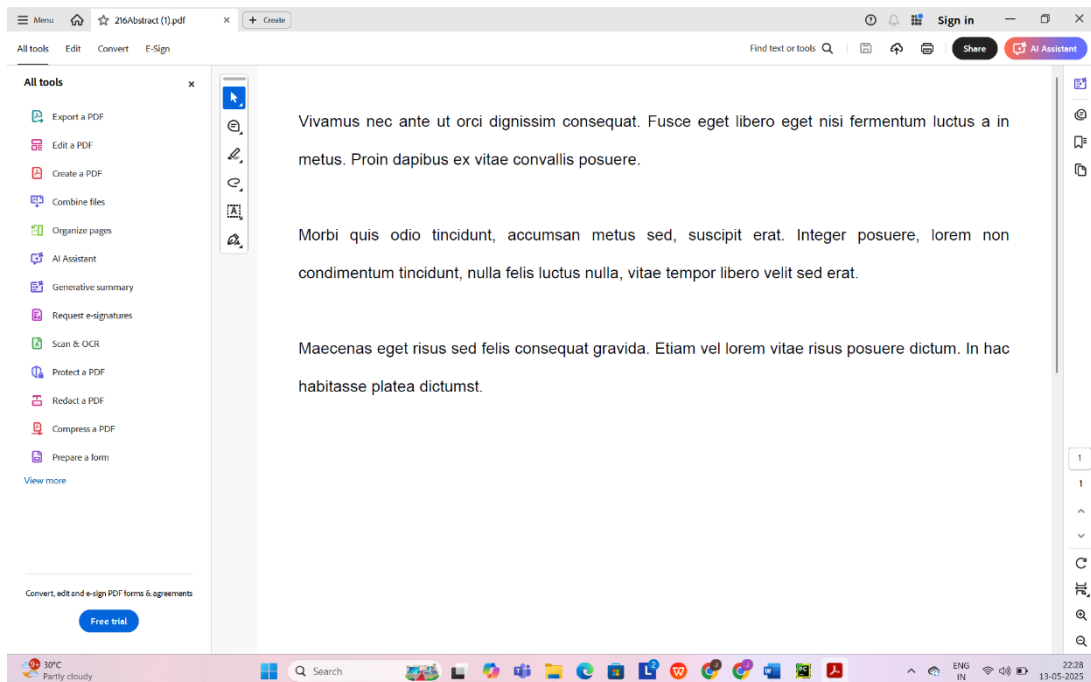


Figure A.9.10: Decoy Document Page

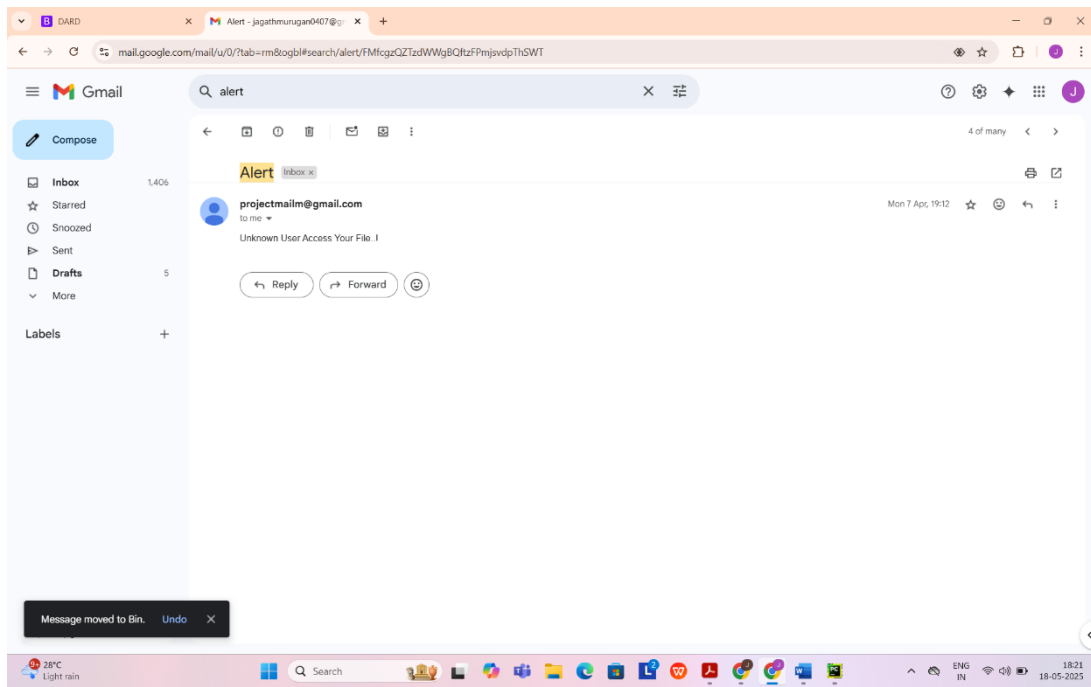


Figure A.9.11: Unauthorized User Access Page

CHAPTER 10

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CHAPTER 11

PUBLICATIONS

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Enhancing Document Security: A Dual Approach with NLP And Cryptographic Techniques

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ABSTRACT

In today's digital era, the risk of document theft and unauthorized access to sensitive information has become increasingly prevalent, posing serious threats to individuals, organizations, and governments. Traditional security measures such as access control and basic encryption often fall short in addressing these challenges, especially with the growing complexity of cyber threats. This project proposes a dual-layered defense system that integrates Natural Language Processing (NLP) and Cryptography to enhance document security. NLP techniques are utilized to analyze, classify, and monitor document content in real-time, enabling the system to identify sensitive data, generate unique document fingerprints, and detect unauthorized changes or content extraction that traditional systems might miss. To strengthen protection further, the system incorporates Elliptic curve cryptography encryption to ensure that documents remain secure during storage and transmission, making them unreadable to unauthorized users. Additionally, the project includes a deception strategy by generating realistic fake documents and storing them in alternative repositories to mislead potential attackers. This hybrid approach of combining content-based monitoring, strong encryption, and decoy deployment creates a comprehensive and proactive solution for securing sensitive information. It offers a significant advancement in combating document theft and unauthorized modification, making it highly applicable across industries that prioritize data confidentiality and integrity.

Keywords: Document Security, Deceptive Repository, Natural Language Processing, Fake Document Storage, Cryptography, Unauthorized Access, Prevention, Elliptic Curve Cryptography

TF-IDF — Term Frequency-Inverse Document Frequency
ECC — Elliptic Curve Cryptography
IP — Intellectual Property
NLP — Natural Language Processing

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